

When does value-directed attention matter?

A normative account of criterion, sensitivity, and decorrelation

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Abstract

When a sensory cue signals that some locations are more valuable than others, an ideal observer can adapt in more than one way: by shifting its decision criteria, by re-allocating spatial attention to sharpen perception where it matters, or—when sensory noise is correlated across locations—by decorrelating the encoded population code. We develop a normative model of value-directed attention in a cued change-detection task that places these three levers within a single optimisation, with the cross-location correlation a free parameter whose zero limit recovers the independent-noise case. Four results follow. Criterion adjustment is typically the dominant route to encoding value, capturing a median of roughly three-quarters of the available reward gain across the parameter space, with a substantial tail in which attention contributes materially. The benefit of attention is non-monotonic in the benefit/cost ratio of enhancement to suppression, peaking in an intermediate band whose lower edge we give in closed form, and it is concentrated in a graded regime—low cue validity, high value contrast, and cost-dominant asymmetry—which we map quantitatively and turn into validity thresholds for cueing designs. Finally, optimal attention never turns away from the high-value location under a predictive cue, a result that holds as a conditional theorem; under a counter-predictive cue, inverted allocation becomes optimal, a new and falsifiable prediction.

1 Introduction

When sensory cues signal that some spatial locations are more rewarding than others, how should an observer allocate limited processing resources? This question is central both to understanding biological attention and to designing the experiments that probe it. An observer informed that a change at one location is worth several times the reward of a change elsewhere could, in principle, exploit that information in more than one way, and the mechanisms differ in what they cost.

The first mechanism is *criterion adjustment*: shifting the decision threshold at the high-value location to become more liberal, accepting more false alarms in exchange for more hits [1, 2]. Criterion adjustment is “free” in the sense that it requires no perceptual reallocation — the observer simply changes how much evidence is needed to declare a change at each location, and a shift at one location leaves sensitivity elsewhere untouched.

The second mechanism is *value-directed attention* (VDA): re-allocating spatial attention toward the high-value location, improving perceptual sensitivity (d') there at the cost of reduced sensitivity elsewhere [3]. Unlike criterion adjustment, VDA involves a genuine perceptual trade-off — resources gained at one location must be taken from another. Reward-associated stimuli can capture spatial attention [4, 5], providing a candidate substrate for such re-allocation.

A third lever becomes available once we abandon the assumption that detection noise is independent across locations. When the population code carries noise that is correlated across locations, an observer can adapt to value by *decorrelating* the encoded signal — and attention is known to reduce interneuronal correlations in visual cortex [3, 6, 7]. We therefore treat the cross-location correlation ρ as a model parameter rather than fixing it at zero, which yields

a three-lever decomposition of the value response — criterion, sensitivity, and decorrelation (Section 2). The familiar criterion-versus-sensitivity account is recovered as the $\rho = 0$ special case.

A further consideration is that the benefit and cost of attention are not necessarily symmetric. Attentional enhancement at attended locations and suppression at unattended locations are dissociable mechanisms that can be modulated independently [8, 9, 10], so a normative model that assumes symmetric benefit and cost may reach conclusions specific to that assumption rather than general properties of the task. We therefore equip the model with an explicit asymmetry ratio $r > 0$ that parameterises the relative efficacy of enhancement versus suppression. The normative question is then: under what conditions does VDA gain meaningfully over an observer that has already optimised its criteria and its validity-driven attention? If the criterion lever captures most of the available reward, the neural machinery supporting value-driven re-allocation is, for this class of tasks, redundant, and experiments designed to observe VDA in that regime will tend to produce null results. Conversely, identifying the parameter regimes in which VDA is essential provides a recipe for experimental designs that can isolate value-driven attentional effects from criterion-based strategies.

Our analysis yields four conclusions (Section 3). First, criterion adjustment is *typically* the dominant lever for encoding value: across a broad parameter sweep the share of the reward gain it captures has a median near 0.76 and is concentrated in [0.30, 1.00], a central tendency with a substantial tail toward attention-dominated outcomes that grows as correlated noise ($\rho > 0$) is admitted. Second, the advantage of VDA is *non-monotonic* in the benefit/cost ratio r , peaking in a cost-dominant regime where value-blind attention is too expensive to deploy; we strengthen this finding with a closed-form threshold $r^\dagger(v)$ that marks the lower edge of the VDA-active band. Third, this advantage is concentrated in a *graded* regime — low cue validity, high value contrast, moderate-to-cost-dominant asymmetry, and low baseline sensitivity — which we map as an iso-VDA contour band over (V, v, r) . Fourth, re-allocating attention *away* from the high-value location is not optimal under predictive cues, a result that holds as a conditional theorem with the explicit condition $V \geq 1/N$; outside that condition, in the counter-predictive “anti-cue” regime ($V < 1/N$), inverted allocation can become optimal, which we report as a new falsifiable prediction of the model. Outside the graded VDA regime — which includes the operating point of many standard spatial-cueing paradigms — an observer that adjusts criteria optimally loses little by treating attention as value-blind.

2 Model

The model is a signal-detection-theoretic account of a cued change-detection task with four nested decision policies. It admits three mechanisms by which an observer can adapt to a value cue: criterion adjustment, sensitivity re-allocation, and decorrelation of cross-location noise (correlation ρ). When $\rho = 0$ the decorrelation lever is inactive and the value response reduces to the familiar criterion-versus-sensitivity description (Section 2.5). We first state the task and the attention-to-perception machinery (Sections 2.1–2.3), then introduce the reward structure and the single place the independence assumption enters it (Section 2.4), and finally the policy decomposition that isolates each lever’s contribution (Section 2.5).

2.1 Task structure

Consider a spatial change-detection task with $N \geq 2$ locations, one of which is cued. Index the locations so that $i = c$ is the cued location and $i = 1, \dots, N - 1$ enumerate the $N - 1$ uncued locations. The cue conveys two pieces of information:

- **Value** ($v \geq 1$): the reward for detecting a change at the cued location. Changes at uncued locations are worth a reward of 1.

- **Validity** ($V \in [1/N, 1]$): the probability that, given a change trial, the change occurs at the cued location. Each uncued location receives probability $(1 - V)/(N - 1)$.

On each trial a change occurs with probability 0.5 (a “change trial”) or does not (a “no-change trial”). Each location carries an equal-variance Gaussian decision variable, and the observer makes a detection decision at each location using signal detection theory. The per-location hit rate and false-alarm rate are

$$\text{HR}_i = \Phi(d'_i/2 - c_i), \quad \text{FAR}_i = \Phi(-d'_i/2 - c_i), \quad (1)$$

where d'_i is perceptual sensitivity at i , c_i is the decision criterion at i , and Φ is the standard-normal cumulative distribution function [1, 2].

2.2 Attention allocation

The observer allocates a continuous share of attention $\alpha_c \in [0, 1]$ to the cued location. The remaining attention is distributed equally among the uncued locations, so each receives $\alpha_u = (1 - \alpha_c)/(N - 1)$ and the shares satisfy $\alpha_c + (N - 1)\alpha_u = 1$. At uniform attention $\alpha_c = 1/N$ and all locations are treated identically.

2.3 Attention-to-perception mapping

Attention determines perceptual sensitivity at each location through a baseline transfer function

$$f(a) = f_0 + (1 - f_0)h(a), \quad (2)$$

where a is the attention allocated to a location, $f_0 \in (0, 1)$ is the baseline sensitivity fraction at minimal attention, and h is a monotonically increasing function with $h(0) = 0$ and $h(1) = 1$. We consider four functional forms,

$$h(a) \in \{a, \sqrt{a}, a^{0.3}, a^2\}, \quad (3)$$

spanning regimes from strongly diminishing returns ($a^{0.3}$) to accelerating returns (a^2). At uniform attention every location achieves a baseline sensitivity $d'_{\text{base}} = d'_{\text{max}} \cdot f(1/N)$.

Independent benefit and cost. Attentional re-allocation need not help and hurt equally. We parameterise the asymmetry between attentional benefit and cost with a ratio $r > 0$. When a location gains attention relative to uniform, its d' departure from baseline is scaled by a benefit weight $\beta(r)$; when a location loses attention, the departure is scaled by a cost weight $\gamma(r)$. Under an additive conservation rule,

$$\beta(r) = \frac{2r}{r+1} \quad (\text{benefit}), \quad \gamma(r) = \frac{2}{r+1} \quad (\text{cost}), \quad (4)$$

which satisfy $\beta + \gamma = 2$ (conserving total magnitude) and $\beta/\gamma = r$. The ratio r reflects the relative efficacy of attentional enhancement versus suppression: a benefit-dominant regime ($r > 1$) corresponds to circuits in which top-down feedback boosts attended representations more effectively than lateral inhibition suppresses unattended ones, and a cost-dominant regime ($r < 1$) to strong surround suppression at unattended locations [8, 9, 10]. The resulting per-location sensitivities are

$$d'_c = d'_{\text{base}} + \beta(r) \cdot [d'_{\text{max}} f(\alpha_c) - d'_{\text{base}}], \quad (5)$$

$$d'_u = d'_{\text{base}} + \gamma(r) \cdot [d'_{\text{max}} f(\alpha_u) - d'_{\text{base}}], \quad (6)$$

with the convention that if $\alpha_c < 1/N$ the roles reverse (the cued departure is scaled by γ and each uncued by β) and all d' values are clamped at ≥ 0 . At $r = 1$, $\beta = \gamma = 1$ and the model

reduces to a symmetric special case in which a single shared transfer function governs both benefit and cost; this symmetric corner is the smooth centre of the asymmetry family, and its exact consistency properties are recorded in the Supplementary material (Section A).

[GAP G-001: needs the attention-to- d' mapping figure.]

2.4 Reward structure and the decorrelation channel

We evaluate two reward variants to separate task-design effects from fundamental incentive structure. *Variant A* (value-coupled correct rejection) couples the correct-rejection reward to value and validity, $\text{CR} = Vv + (1 - V)$, calibrated so that the expected value of responding and withholding are balanced at uniform attention. *Variant B* (fixed correct rejection) sets $\text{CR} = 1$ independent of v and V , which creates a stronger marginal benefit for detection at high-value locations because the hit-CR difference at the cued position scales with v . The expected reward on a single trial is

$$\mathbb{E}[R] = \frac{1}{2} [V \text{HR}_c v + (1 - V) \text{HR}_u] + \frac{1}{2} P_{\text{no-fa}} \text{CR}, \quad (7)$$

where $P_{\text{no-fa}}$ is the probability of no false alarm at any location on a no-change trial.

The locus of the independence assumption. Equation (7) is linear in the marginal hit rates HR_c, HR_u : on a change trial the change occurs at a single location, so no product of probabilities across locations appears in the change-trial bracket. The *only* cross-location term in (7) is $P_{\text{no-fa}}$. Under independent per-location decision variables,

$$P_{\text{no-fa}}^{\text{indep}} = \prod_i (1 - \text{FAR}_i) = \Phi(b_c) \Phi(b_u)^{N-1}, \quad b_i := c_i + d'_i/2, \quad (8)$$

where b_i is the z -score the no-change decision variable must stay below. Equation (8) is the sole place the independence assumption enters the reward structure, so relaxing that assumption amounts to replacing this product by the correlated joint orthant probability.

The decorrelation channel. We promote the independence assumption to a tunable equicorrelation parameter $\rho \in [0, 1]$. Let the no-change decision variables be jointly Gaussian with standardised marginals and equicorrelation,

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma), \quad \mu_i = -d'_i/2, \quad \Sigma_{ii} = 1, \quad \Sigma_{ij} = \rho \ (i \neq j). \quad (9)$$

The one-factor representation $X_i = \mu_i + \sqrt{\rho} Z + \sqrt{1 - \rho} \varepsilon_i$, with $Z, \varepsilon_1, \dots, \varepsilon_N$ i.i.d. $\mathcal{N}(0, 1)$, reproduces (9) and makes the X_i conditionally independent given Z . Multiplying the conditional event probabilities and integrating out Z against the standard-normal density φ yields the exact one-dimensional integral

$$P_{\text{no-fa}}(\rho) = \int_{-\infty}^{\infty} \Phi\left(\frac{b_c - \sqrt{\rho} z}{\sqrt{1 - \rho}}\right) \Phi\left(\frac{b_u - \sqrt{\rho} z}{\sqrt{1 - \rho}}\right)^{N-1} \varphi(z) dz. \quad (10)$$

Equation (10) is exact for any N and any $\rho \in [0, 1]$; equicorrelation collapses the N -dimensional orthant integral to a one-dimensional one, with no multivariate-normal CDF required. We evaluate it by 64-node Gauss-Hermite quadrature, with quadrature error $\leq 10^{-15}$ against a 128-node reference across the operating band $\rho \in \{0, 0.05, 0.1, 0.2, 0.3, 0.4\}$.

Recovery contract. At $\rho = 0$ the integrand of (10) is constant in z and the integral reduces to (8):

$$P_{\text{no-fa}}(0) = \Phi(b_c) \Phi(b_u)^{N-1}. \quad (11)$$

Equation (11) is the *recovery contract*: with ρ clamped to zero, the model’s reward, VDA, and criterion-fraction values coincide to floating-point identity with the independent-noise model, so every result reported below contains the independent-noise case as its $\rho = 0$ limit. The empirical envelope we explore is $\rho \in [0, 0.4]$, which brackets the $\rho \approx 0.2$ spike-count correlation reported in macaque V4 during peripheral-cue change detection [3]; higher ρ values are admissible in the model but anchor no headline claim. Equicorrelation is a deliberate one-parameter simplification; structured covariances [6, 7] break the exact one-dimensional reduction (10) and are noted as a scoped limitation in the Discussion (Section 4). By the Slepian inequality, $P_{\text{no-fa}}(\rho)$ is non-decreasing in ρ at fixed (b_c, b_u) [11]; the full derivation of (10) and this monotonicity is in the Supplementary material (Section A).

2.5 Policy decomposition

To isolate the contribution of each mechanism, we define four nested policies that optimise (7) over progressively shrinking parameter sets:

1. **Joint optimum (P1):** α_c, c_c, c_u jointly optimised at each v . This is the best any observer can achieve.
2. **Value-blind attention (P2):** α_c fixed at its $v = 1$ optimum — the optimal allocation when all locations have equal value — with the criteria re-optimised at each v . This observer uses attention for validity but not for value.
3. **Uniform attention (P3):** $\alpha_c = 1/N$, criteria optimised. This observer uses only the criterion to encode value.
4. **Floor (P4):** $\alpha_c = 1/N, c = 0$. An unbiased baseline that uses neither mechanism.

The reward gain decomposes into three additive pieces,

$$\text{criterion gain} = R_{\text{P3}} - R_{\text{P4}}, \quad (12)$$

$$\text{validity-attention gain} = R_{\text{P2}} - R_{\text{P3}}, \quad (13)$$

$$\text{VDA} = R_{\text{P1}} - R_{\text{P2}}, \quad (14)$$

where the value-directed-attention benefit VDA — the reward gained by making attention track cue *value* on top of an already optimal, validity-driven allocation and criterion — is our central metric. The criterion fraction,

$$\text{CF} = \frac{R_{\text{P3}} - R_{\text{P4}}}{R_{\text{P1}} - R_{\text{P4}}}, \quad (15)$$

measures the share of the total reward gain captured by the criterion lever alone. We report the range and distribution of CF across the full parameter space in Section 3.

Three levers. At fixed correlation the value response decomposes into two levers: criterion shift ($\text{P3} \rightarrow \text{P4}$) and sensitivity re-allocation ($\text{P1} \rightarrow \text{P2}$). With ρ a free model parameter, the same expected-reward expression (7) admits a third.

Definition 1 (The three levers). At fixed $(N, V, v, d'_{\text{max}}, f_0, h, \text{CR}, r)$, the model exposes three independent mechanisms by which an observer can adapt to a non-trivial cue:

1. **Criterion shift** — moving (c_c, c_u) off the $v = 1$ floor. Its standalone reward contribution is $R_{\text{P3}} - R_{\text{P4}}$, and its share of the value-related gain is CF (15).

2. **Sensitivity re-allocation** — moving α_c off the $v = 1$ optimum. Its standalone contribution at adapted criteria is $R_{P1} - R_{P2} = \text{VDA}$ (14).
3. **Decorrelation** — moving ρ off the $\rho = 0$ corner. By Slepian monotonicity $P_{\text{no-fa}}(\rho)$ is non-decreasing in ρ at fixed (b_c, b_u) [11], which lifts every per-policy supremum reward $R_{P_k}(\rho)$ above its $\rho = 0$ value.

The decorrelation lever is grounded in physiology: attention is known to reduce interneuronal correlations in visual cortex [3, 6, 7]. Treating ρ as a model parameter rather than fixing it at zero lets us report what the expected-reward expression does as cross-location noise is varied. Whether independent-noise analyses bound the VDA benefit, and what quantity they bound, is an empirical question about the sign of $\partial \text{VDA} / \partial \rho$ that we take up with the corresponding sweep in the Results (Section 3) and Discussion (Section 4).

3 Results

We present four findings that develop a single argument about when value-directed attention (VDA) matters. First, across the parameter space the criterion lever *typically* captures the bulk of the value-related reward gain, as a central tendency with a genuine tail toward attention-dominated outcomes (Section 3.1). Second, the residual VDA benefit is *non-monotonic* in the benefit/cost ratio r , rising out of zero, peaking, and decaying again, with a closed-form escape threshold marking the lower edge of the active band. Third, the region of (V, v, r) space in which VDA is materially large is concentrated, with a *graded* boundary that we map as a contour band. Fourth, optimal attention does not invert under a predictive cue — a *conditional* theorem (condition $V \geq 1/N$), with inversion in the anti-cue corner $V < 1/N$ as a new falsifiable prediction. Throughout, the decorrelation channel ρ (Section 2) is reported *alongside* each finding as a sensitivity: this is the operational content of the three-lever decomposition (Definition 1).

3.1 Criterion adjustment typically dominates value encoding

The first finding concerns how the optimal observer divides the value-related reward gain between the two adaptive levers available at fixed correlation: re-aiming the per-location decision criterion (the criterion shift, $R_{P3} \rightarrow R_{P4}$) and re-allocating perceptual sensitivity (VDA, $R_{P1} \rightarrow R_{P2}$). The criterion fraction CF (Eq. 15) is the share of the total value-related gain captured by the criterion lever alone. Across the primary 4,410-cell (r, V, v) sweep the criterion lever *typically* dominates value encoding — but as a distribution with a central tendency near three-quarters and a substantial tail, rather than a uniform bound.

The criterion fraction is a distribution. Evaluating the model at the headline configuration $(N, d'_{\text{max}}, f_0, h) = (4, 2, 0.5, \sqrt{\cdot})$ across the primary sweep ($V \in [0.25, 1.00]$, 21 points; $r \in [0.1, 10]$, 22 values — 21 log-spaced plus a pinned $r = 1$; $v \in \{1, 2, 3, 4, 5\}$; both reward variants), of whose 4,620 nominal cells 4,410 (2,205 per variant) yield valid, non-degenerate optima, gives the CF distribution of Table 1. The median CF is 0.7552 (variant A) and 0.7682 (variant B), well above 0.5, while the distribution spans a wide range at both ends. Variant A’s strict minimum is 0.5587; variant B’s is 0.3040, with 8% of variant-B cells falling under 0.50. At the upper end both variants reach $\text{CF} = 1.00$. The finding is therefore a *central-tendency* one: the criterion lever captures roughly three-quarters of the value-related reward gain in a typical cell, with a left tail — much heavier in variant B — in which it captures substantially less. Figure 1 shows the full per-cell distribution.

variant	CF _{min}	q _{5%}	q _{25%}	median	q _{75%}	q _{95%}	CF _{max}	mean $\pm$ s.d.
A	0.5587	0.5931	0.6530	0.7552	0.8996	1.0000	1.0000	0.7773 $\pm$ 0.1363
B	0.3040	0.4564	0.6288	0.7682	0.9463	1.0000	1.0000	0.7691 $\pm$ 0.1808

Table 1: Distribution of the criterion fraction CF (Eq. 15) across the primary 4,410-cell (r, V, v) sweep at the headline configuration $(N, d'_{\max}, f_0, h) = (4, 2, 0.5, \sqrt{\cdot})$ under independence $\rho = 0$, per reward variant. Quantiles are over the $n = 2,205$ valid cells per variant. The median (boldface) sits near three-quarters in both variants, with a wide spread toward lower CF.

variant	r -regime (V)	median CF	strict min	frac < 0.6	n
A	cost-dom., V low	0.998	0.825	0.000	350
A	cost-dom., V high	0.897	0.825	0.000	385
A	symmetric, V low	0.779	0.665	0.000	350
A	symmetric, V high	0.740	0.665	0.000	385
A	benefit-dom., V high	0.645	0.563	0.062	385
A	benefit-dom., V low	0.612	0.559	0.374	350
B	cost-dom., V low	1.000	0.905	0.000	350
B	cost-dom., V high	0.931	0.797	0.000	385
B	symmetric, V low	0.800	0.497	0.103	350
B	symmetric, V high	0.753	0.575	0.010	385
B	benefit-dom., V high	0.630	0.469	0.317	385
B	benefit-dom., V low	0.509	0.304	0.780	350

Table 2: Quadrant breakdown of CF at $\rho = 0$. The criterion lever dominates uniformly in the cost-dominant region (median CF ≥ 0.90 in both variants) and cedes to attention re-allocation in the benefit-dominant, low-validity corner, where its median falls to 0.61 (variant A) and 0.51 (variant B).

Where the criterion lever dominates, and where it cedes. The marginal distribution obscures the *regime* structure that drives it. Partitioning the cells by benefit/cost regime ($r < 1$ cost-dominant, $r = 1$ symmetric, $r > 1$ benefit-dominant) and validity ($V < 1/2$ vs. $V \geq 1/2$) exposes a clean ordering (Table 2): CF decreases monotonically along the r axis from the cost-dominant corner (median CF ≥ 0.90 in both variants — criterion captures almost everything) to the benefit-dominant, low-validity corner, where it cedes the lead to attention re-allocation (variant A median 0.61 with 37% of cells below 0.6; variant B median 0.51 with 78% below 0.6 and a strict minimum of 0.30). Read off the quadrant medians, the ordering runs from CF near unity in the cost-dominant regime, through roughly three-quarters at symmetry, to a benefit-dominant, low-validity corner in which the criterion lever is no longer dominant. Figure 2 displays the same structure cell-wise over the (r, V) plane, and Figure 3 shows $CF(r)$ declining from near unity as the benefit/cost ratio rises. The criterion lever thus tends to unity at small r and declines at large r , with the share it captures spread over a wide range across the parameter space.

Decorrelation sensitivity. Admitting correlated noise reshapes the criterion fraction in a variant-specific way. Running the full sweep at $\rho = 0.2$ — the cross-location correlation magnitude reported for visual cortex [3] — the criterion fraction falls in variant A. Raising ρ from 0 to 0.2 amplifies the variant-A left tail by roughly a factor of three (the fraction of cells with $CF < 0.6$ rises from 0.07 to 0.22) and pushes the strict minimum below 0.50; cell-wise, CF decreases in 84% of variant-A cells (median $\Delta CF = -0.035$). In variant B the response is mixed: 64% of cells decrease while 24% increase, consistent with an essentially flat variant-B sensitivity. The decorrelation channel thus reduces CF on a typical variant-A cell and leaves it essentially

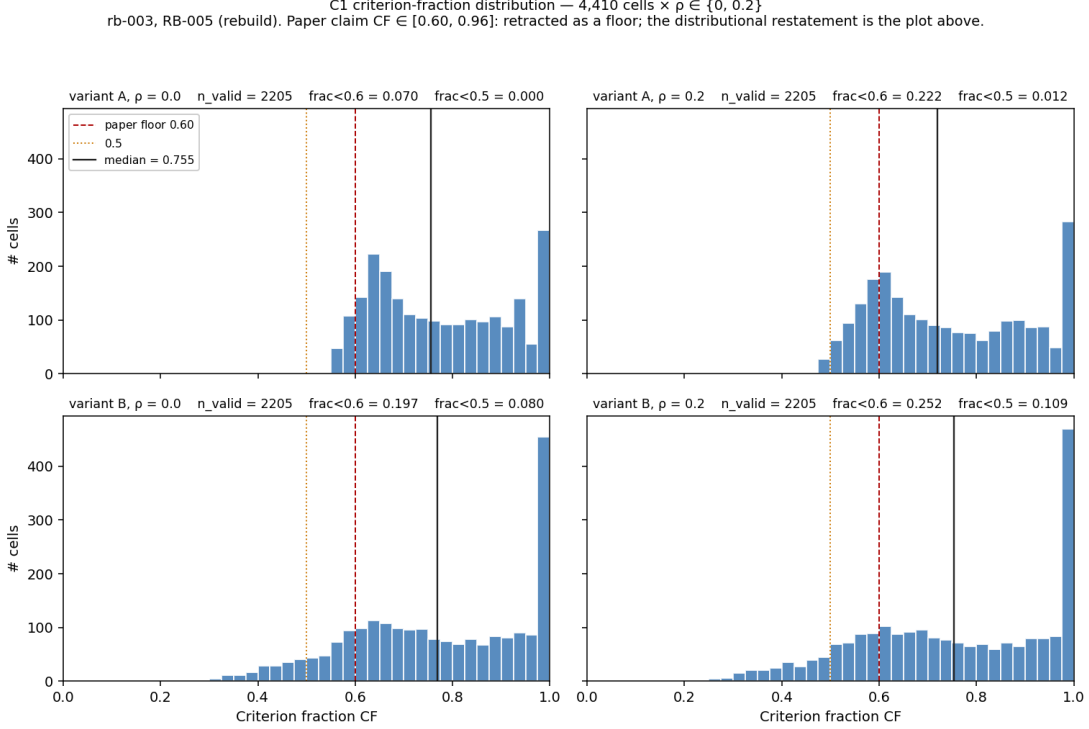


Figure 1: Per-cell histogram of the criterion fraction CF across the 4,410-cell sweep (columns: reward variant; rows: decorrelation $\rho \in \{0, 0.2\}$). The distribution — a broad mode near $CF \approx 0.65$, a left tail (much heavier in variant B), and a spike at $CF = 1$ — characterises how the criterion fraction is spread across the parameter space. The $\rho = 0.2$ row exhibits the decorrelation amplification of the left tail quantified below.

unchanged, with substantial scatter, on a typical variant-B cell. The sign of $\partial VDA / \partial \rho$ is itself non-uniform and is taken up with the VDA findings below and in the Discussion (Section 4).

Robustness to the conservation rule. The benefit and cost weights (β, γ) satisfy a conservation constraint (Section 2); the numbers above are reported at its additive endpoint ($\beta + \gamma = 2$), one member of a one-parameter family whose opposite endpoint is multiplicative ($\beta \cdot \gamma = 1$). Sweeping that family at each reward variant, the central tendency of CF is robust while the tail is not. Moving from the additive to the multiplicative endpoint leaves the median essentially fixed (variant A -0.0012 ; variant B -0.0042) but roughly doubles the fraction of cells with $CF < 0.5$ (from 4.0% to 8.3%, combined), drives 191 cells from $CF \geq 0.5$ to $CF < 0.5$ with no cell moving the other way, and deepens the variant-B minimum from 0.304 to 0.231. The conservation-family treatment of these band endpoints is given in the Discussion (Section 4); for the present finding the conclusion is that the “criterion typically dominates” central tendency is invariant to the conservation rule, whereas the size of the tail in which it does not dominate is rule-dependent.

3.2 The value-directed attention benefit is non-monotonic in the benefit/cost ratio

The first finding fixed the cross-location correlation and asked how the optimal observer splits the value-related gain between the criterion and attention levers. We now follow the attention lever itself across the benefit/cost ratio r and find that its contribution is *non-monotonic*: the value-directed attention benefit $VDA(r) = R_{P1}(r) - R_{P2}(r)$ (Eq. 14) rises out of zero at small r ,

C1 criterion-fraction heatmap, $CF(r, V)$ at $v = 5$
rb-003, RB-005 — distribution structure, not a categorical floor.

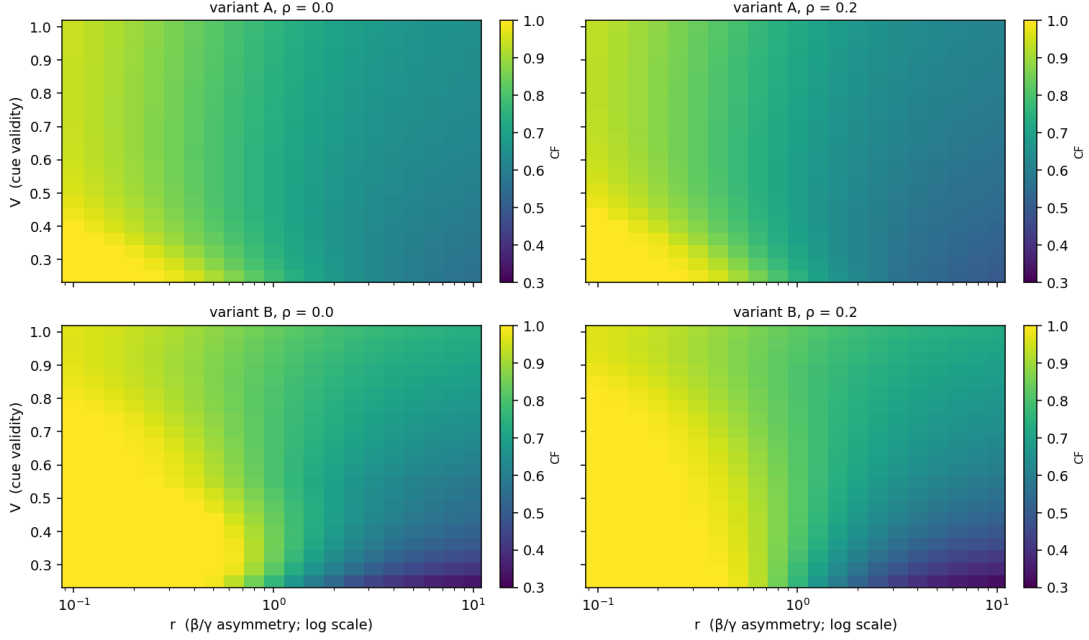


Figure 2: CF over the (r, V) plane at $v = 5$, $(N, d'_{\max}, f_0, h) = (4, 2, 0.5, \sqrt{\cdot})$ (rows: ρ ; columns: variant). The diagonal boundary in $(\log r, V)$ separates the criterion-dominated region (CF near 1, low r) from the attention-dominated corner (CF low, high r , low V) of Table 2. The low-CF corner is wider in variant B and widens further at $\rho = 0.2$.

reaches an interior maximum, and decays back toward zero as $r \rightarrow \infty$. This shape is the model’s signature: attention re-allocation pays off only in an intermediate band of the benefit/cost ratio, and the model delivers a closed-form expression for the lower edge of that band.

A closed-form escape threshold. The asymmetry between the reward for a benefit (correctly attending the high-value cued location) and the cost (a miss or a false alarm) governs when re-allocating sensitivity away from uniform attention first becomes worthwhile. Differentiating the joint-optimum expected reward $R_{P1}(\alpha, r)$ with respect to the cued-attention share α and evaluating at the uniform allocation $\alpha = 1/N$, the first-order condition $\partial_{\alpha} R_{P1}|_{1/N} = 0$ defines the value of the benefit/cost ratio at which the optimiser begins to move attention toward the cued location. Writing $d'_b \equiv d'_{\text{base}} = d'_{\max} f(1/N)$ for the baseline discriminability under uniform attention (Section 2) and (c_c^*, c_u^*) for the per-location optimal criterion pair at $\alpha = 1/N$, this escape threshold is

$$r^{\dagger}(v) = \frac{K_u(v)}{(N-1) \cdot K_c(v)}, \quad (16)$$

with per-location gradient coefficients

$$K_c(v) = \frac{1}{4} \left[V v \varphi\left(\frac{d'_b}{2} - c_c^*\right) + \text{CR}(v) \varphi\left(\frac{d'_b}{2} + c_c^*\right) \Phi\left(\frac{d'_b}{2} + c_u^*\right)^{N-1} \right], \quad (17)$$

$$K_u(v) = \frac{1}{4} \left[(1-V) \varphi\left(\frac{d'_b}{2} - c_u^*\right) + (N-1) \text{CR}(v) \Phi\left(\frac{d'_b}{2} + c_c^*\right) \cdot \Phi\left(\frac{d'_b}{2} + c_u^*\right)^{N-2} \varphi\left(\frac{d'_b}{2} + c_u^*\right) \right]. \quad (18)$$

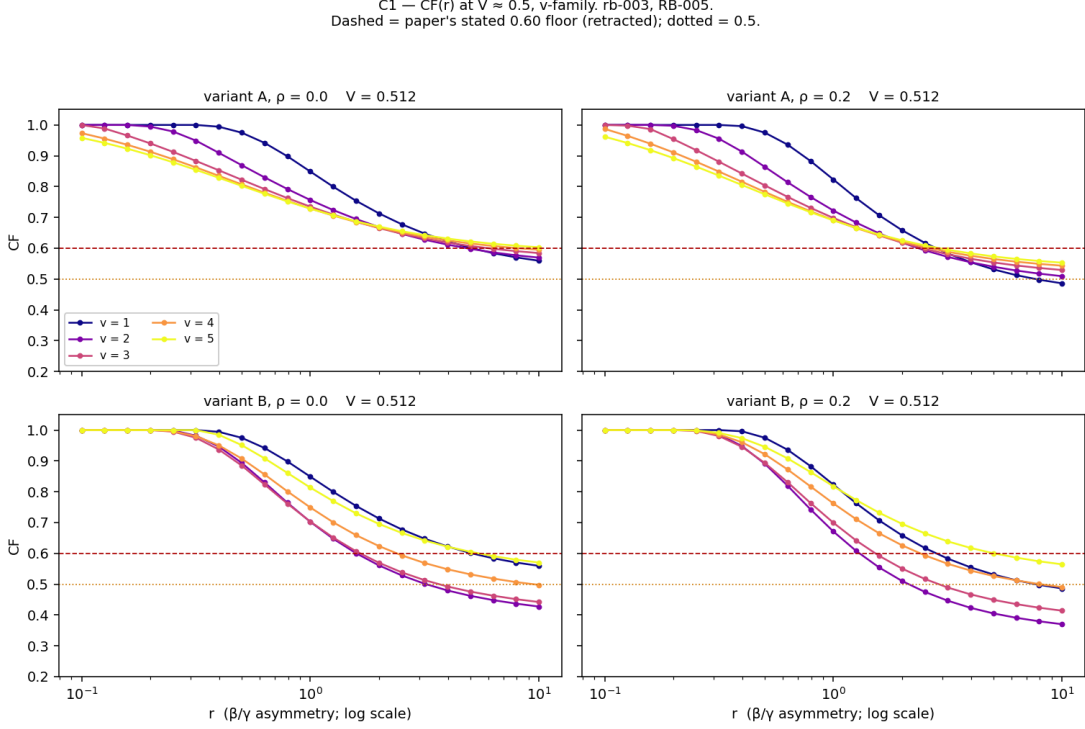


Figure 3: $CF(r)$ at $V \approx 0.5$ as a v -family $v \in \{1, 2, 3, 4, 5\}$ (rows: ρ ; columns: variant). At small r (cost-dominant) the curves cluster near $CF = 1$; as r increases they decline, more steeply for larger v , exposing the benefit-dominant region in which attention re-allocation overtakes the criterion shift.

The factor $CR(v)$ is the correct-rejection reward scaling set by the reward variant (Section 2); the value-blind reference used throughout this section takes $CR(v) = 1$, with the benefit and cost weights following the additive conservation rule ($\beta + \gamma = 2$). The threshold (16) is a consequence of the model definitions rather than an empirical regularity; its derivation is given in the Supplementary material (Section A).

The threshold falls as cued value rises. Evaluating Eqs. (16)–(18) at the configuration $(N, d'_{\max}, f_0, h, V) = (4, 2, 0.5, \sqrt{\cdot}, 0.5)$ gives the values in Table 3. The escape threshold decreases monotonically in the cued value v , from $r^\dagger \approx 0.34$ at $v = 1$ to $r^\dagger \approx 0.02$ at $v = 10$: the more reward a correct cued response carries, the smaller the benefit/cost asymmetry needed before attention re-allocation begins to pay. Figure 4 plots $r^\dagger(v)$ as a continuous curve with these family points marked.

The peak benefit sits inside the predicted band. The escape threshold marks where the joint optimum P1 *begins* to depart from uniform attention. The value-blind baseline P2, against which VDA is measured, leaves uniform attention only at the higher threshold $r^\dagger(v=1)$ — the symmetric reference, because P2 allocates attention as if every location carried equal value. The VDA benefit therefore opens up on the interior interval $(r^\dagger(v), r^\dagger(1))$, where P1 has re-allocated but P2 has not, and closes again once P2 also escapes uniform attention. This predicts that the peak of $VDA(r)$ lies strictly above $r^\dagger(v)$. A high-resolution sweep of 84 points in r (83 log-spaced over $[0.1, 10]$ plus pinned escape- and peak-neighbourhood points) confirms it: the peak $r^* = \arg \max_r VDA(r)$ exceeds $r^\dagger(v)$ for every v in the family, by a margin of 0.28 to 0.35, and clusters just below the symmetric anchor $r^\dagger(1) \approx 0.34$ (Table 4).

Figure 5 shows the full $VDA(r)$ family. The peak height grows monotonically with cued

v	$r^\dagger(v)$	c_c^*	c_u^*	$K_c(v)$	$K_u(v)$
1	0.343	0.40	1.05	0.0930	0.0958
2	0.168	0.25	1.30	0.1734	0.0872
3	0.099	0.15	1.50	0.2532	0.0755
5	0.050	0.10	1.75	0.4065	0.0615
8	0.022	0.05	2.05	0.6357	0.0424
10	0.016	0.05	2.15	0.7864	0.0380

Table 3: Closed-form escape threshold $r^\dagger(v) = K_u(v)/[(N-1)K_c(v)]$ from (16)–(18) at $(N, d'_{\max}, f_0, h, V) = (4, 2, 0.5, \sqrt{\cdot}, 0.5)$, additive conservation. The $v = 1$ row is the symmetric reference $r^\dagger(1) \approx 0.343$; the threshold falls monotonically as the cued value grows.

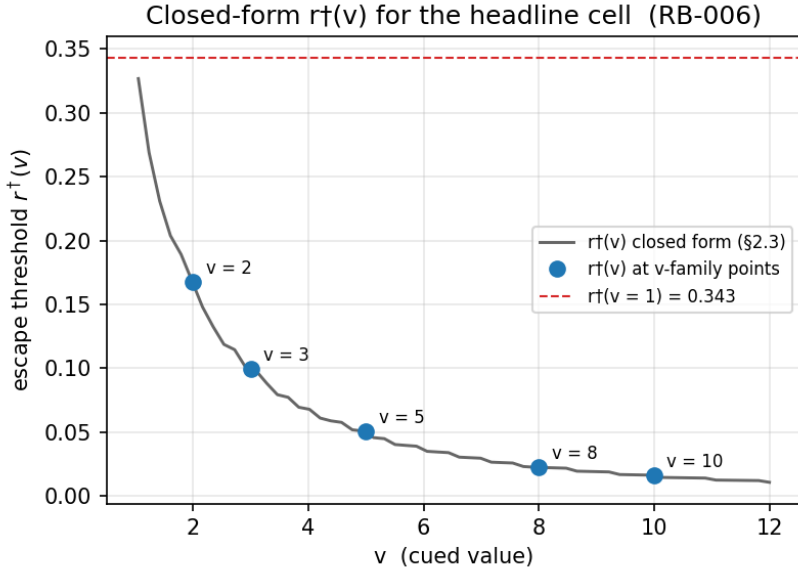


Figure 4: The closed-form escape threshold $r^\dagger(v)$ at $(N, d'_{\max}, f_0, h, V) = (4, 2, 0.5, \sqrt{\cdot}, 0.5)$, additive conservation. The continuous curve is computed from (16)–(18) on a dense v -grid, with the family points $v \in \{2, 3, 5, 8, 10\}$ marked. The horizontal reference $r^\dagger(1) \approx 0.343$ is the symmetric upper anchor of the family. The threshold marks the benefit/cost ratio at which the optimal observer first leaves uniform attention.

value, from VDA ≈ 0.012 at $v = 2$ to VDA ≈ 0.183 at $v = 10$, and each peak sits above its corresponding threshold $r^\dagger(v)$ (dashed verticals). The benefit of value-directed attention is thus largest precisely where the cued location is most valuable and the benefit/cost asymmetry is moderate — not at the extremes of r , where either the criterion lever alone suffices (small r) or both policies have already re-allocated (large r).

Decorrelation sensitivity. The sign of the decorrelation lever’s effect on the VDA benefit depends on the cued value, not on the benefit/cost ratio alone. Raising the cross-location correlation from $\rho = 0$ to $\rho = 0.2$ — the magnitude reported for visual cortex [3] — and re-measuring the peak benefit across the v -family (Table 5) *suppresses* the peak for $v \leq 8$ (by up to $\Delta\text{VDA}^* = -0.004$ at $v = 3$) but *amplifies* it at $v = 10$ ($\Delta\text{VDA}^* = +0.001$). The peak location r^* also drifts upward in ρ , most visibly at low v , where it rises from 0.501 to 0.631 at $v = 2$. This upward drift has a closed-form counterpart: extending the threshold (16)–(18) to $\rho > 0$ by replacing the no-false-alarm terms in K_c, K_u with their correlation-aware one-dimensional Gauss–Hermite gradients (Section A) yields $r^\dagger(v; \rho)$, and predicts $r^\dagger(v; 0.2) > r^\dagger(v; 0)$ at every

v	$r^\dagger(v)$	r^*	$r^* - r^\dagger(v)$	above?
2	0.168	0.501	+0.334	✓
3	0.099	0.376	+0.276	✓
5	0.050	0.376	+0.325	✓
8	0.022	0.376	+0.354	✓
10	0.016	0.355	+0.339	✓

Table 4: Empirical peak $r^* = \arg \max_r \text{VDA}(r)$ against the closed-form escape threshold $r^\dagger(v)$ at $(N, d'_{\max}, f_0, h, V) = (4, 2, 0.5, \sqrt{\cdot}, 0.5)$, $\rho = 0$. The peak lies above $r^\dagger(v)$ for every v and clusters near the symmetric anchor $r^\dagger(1) \approx 0.343$, consistent with the prediction that VDA is supported on the interval $(r^\dagger(v), r^\dagger(1))$. Peak located on an 84-point log-grid in r .

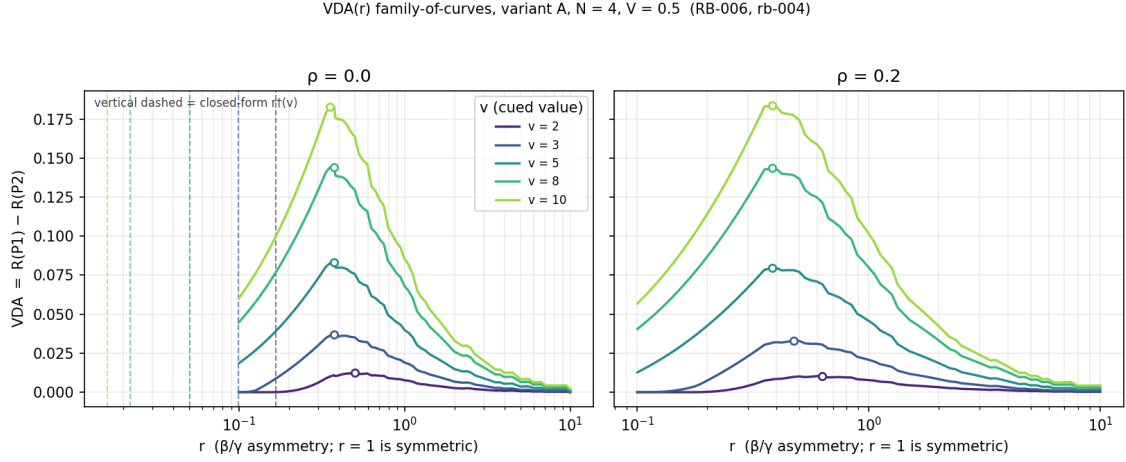


Figure 5: Non-monotonicity of $\text{VDA}(r)$ in the benefit/cost ratio across a v -family at $(N, d'_{\max}, f_0, h, V) = (4, 2, 0.5, \sqrt{\cdot}, 0.5)$, additive conservation. **Left:** $\rho = 0$. Each curve rises from zero, peaks in the interior, and decays; the closed-form escape threshold $r^\dagger(v)$ from (16) is overlaid as a vertical dashed line per v , and the empirical peak (white-filled dot) lies above $r^\dagger(v)$ for every v (Table 4). **Right:** the same family at $\rho = 0.2$; correlated noise suppresses the peak at low v but amplifies it at $v = 10$, so the sign of $\partial \text{VDA}^* / \partial \rho$ depends on the cued value (Table 5).

v — the lower edge of the active band drifts up by +3% at $v = 1$ to +30% at $v = 8$, matching the sign of the empirical peak drift at every $v \neq 1$. The decorrelation lever therefore does not uniformly attenuate the VDA benefit: in the strongly benefit-dominant, high-value corner it *enhances* the benefit, and across the family it shifts the active band toward larger r . This is the operational meaning of treating decorrelation as a third independent lever (Definition 1): its effect on attention is signed by the value structure of the task.

Scope. The threshold and family numerics above are reported under additive conservation ($\beta + \gamma = 2$, $\text{CR}(v) = 1$) at $V = 0.5$; the correlation-aware extension of (16)–(18) and its drift table are developed in the Supplementary material (Section A). The sensitivity of the peak magnitudes to the conservation rule is taken up with the conservation-family band in the Discussion (Section 4).

3.3 The benefit is concentrated in a graded regime

The non-monotonicity of Section 3.2 locates the active band along the r -axis at a fixed design point. Sweeping the experimental-design plane (V, v) shows that the value-directed attention

v	VDA * ($\rho=0$)	VDA * ($\rho=0.2$)	Δ VDA *	r^* ($\rho=0$)	r^* ($\rho=0.2$)
2	0.01233	0.01036	-0.00197	0.501	0.631
3	0.03698	0.03291	-0.00407	0.376	0.473
5	0.08300	0.07954	-0.00345	0.376	0.383
8	0.14422	0.14355	-0.00067	0.376	0.383
10	0.18284	0.18387	+0.00103	0.355	0.383

Table 5: Decorrelation sensitivity of the peak VDA benefit at $(N, d'_{\max}, f_0, h, V) = (4, 2, 0.5, \sqrt{\cdot}, 0.5)$, additive conservation. Raising ρ from 0 to 0.2 suppresses peak VDA * for $v \leq 8$ and amplifies it at $v = 10$, while the peak location r^* drifts upward (most visibly at low v). The sign of $\partial \text{VDA}^* / \partial \rho$ thus depends on the cued value.

r	ρ	VDA $_{\min}$	median	$q_{95\%}$	VDA $_{\max}$	frac ≥ 0.005	frac ≥ 0.05
0.3	0.0	0	0.0010	0.1289	0.1734	0.465	0.287
0.3	0.2	0	0.0040	0.1328	0.1780	0.484	0.295
1.0	0.0	0	0.0052	0.1152	0.1574	0.503	0.219
1.0	0.2	0	0.0069	0.1180	0.1552	0.535	0.236
3.0	0.0	0	0.0024	0.0361	0.0619	0.380	0.012
3.0	0.2	0	0.0025	0.0390	0.0621	0.402	0.019

Table 6: Per-panel distribution of VDA over the $31 \times 19 = 589$ (V, v) cells, at the headline configuration $(N, d'_{\max}, f_0, h) = (4, 2, 0.5, \sqrt{\cdot})$, variant A. The median VDA ≤ 0.007 in every panel: most of the design plane carries near-zero benefit. The frac ≥ 0.05 column collapses from $\approx 29\%$ at $r = 0.3$ to $\leq 2\%$ at $r = 3$, quantifying the flattening of the contour band along the r -axis.

benefit is materially large only in a concentrated corner of the design space — low cue validity, high value contrast, and moderate-to-low benefit/cost asymmetry — and that this corner has a *graded* boundary. We map the boundary as an iso-VDA contour band and use it to derive a quantitative, threshold-indexed prediction for spatial cueing designs. As elsewhere, the decorrelation channel ρ (Section 2) is reported alongside the map as a sensitivity.

An iso-VDA contour band over the design plane. We evaluated the model at the headline configuration $(N, d'_{\max}, f_0, h) = (4, 2, 0.5, \sqrt{\cdot})$ under variant A on a grid of $V \in [0.25, 1.00]$ (31 uniform points, step 0.025), $v \in [1.0, 10.0]$ (19 uniform points, step 0.5), $r \in \{0.3, 1, 3\}$ (cost-dominant, symmetric, and benefit-dominant asymmetry strata), and $\rho \in \{0, 0.2\}$ (independence plus the Cohen–Maunsell [3] $r_{SC} \approx 0.2$ cortical anchor): $31 \times 19 \times 3 \times 2 = 3,534$ design cells. Each cell returns the optimal allocation, criteria, and the nested-policy rewards from which VDA is formed. Figure 6 presents the iso-VDA contour band as a 2×3 panel grid (rows: ρ ; columns: r). In every panel the bulk of (V, v) space carries near-zero VDA (median VDA ≤ 0.007 ; Table 6), and substantial benefit is confined to a corner along low V and high v . The corner flattens along the r -axis: at the cost-dominant column $r = 0.3$ it is broad and reaches VDA ≈ 0.17 ; at the symmetric column $r = 1$ it narrows to VDA ≈ 0.16 ; at the benefit-dominant column $r = 3$ it collapses to VDA ≤ 0.06 everywhere on the grid.

Distribution of the benefit across the plane. Table 6 summarises the per-panel distribution of VDA over the $589 = 31 \times 19$ design cells in each panel. The peak VDA over the grid is 0.173 at $(r, \rho) = (0.3, 0)$ and falls to 0.062 at $(3.0, 0)$; the fraction of cells with VDA ≥ 0.05 falls from 28.7% at $r = 0.3$ to 1.2% at $r = 3$. At $\rho = 0.2$ this fraction is slightly larger in every panel (+0.8 to +1.7 percentage points), consistent with the ρ -sensitivity below; the peak and median values are essentially unchanged in ρ at this magnitude.

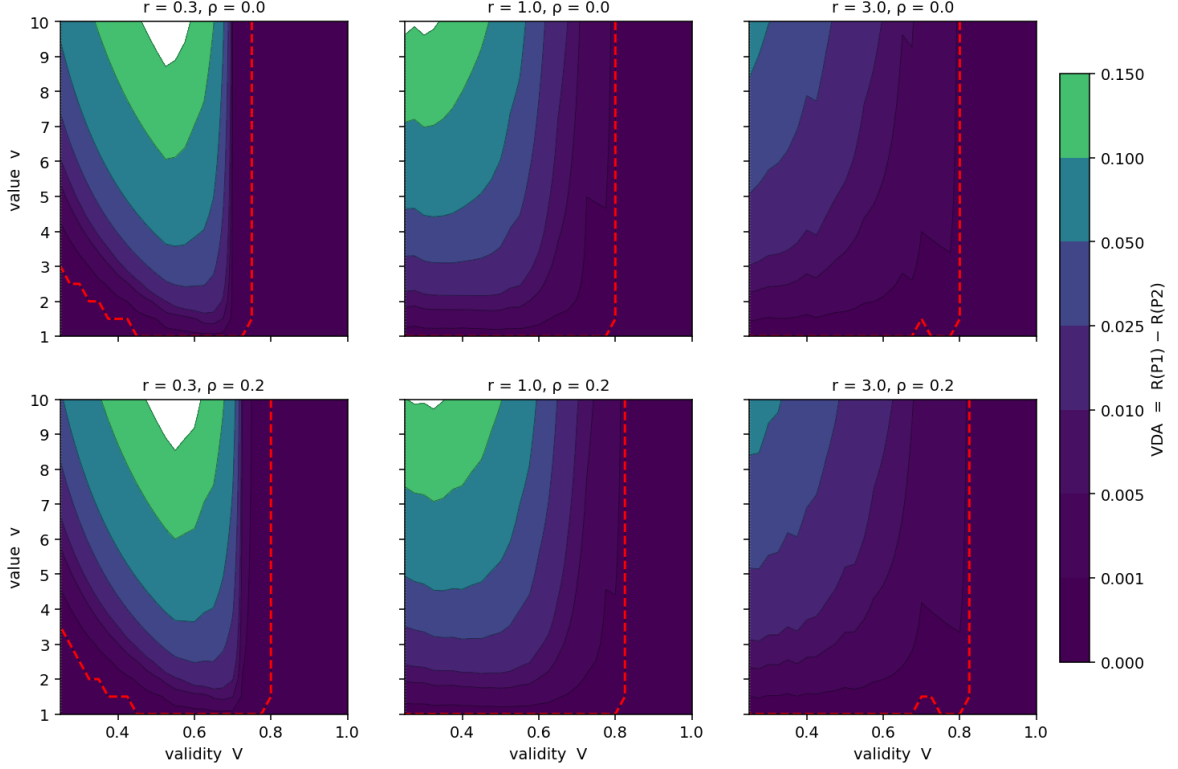


Figure 6: Iso-VDA contour band over the experimental-design plane (V, v) at the headline configuration $(N, d'_{\max}, f_0, h) = (4, 2, 0.5, \sqrt{ })$, variant A. Rows: cross-location correlation $\rho \in \{0, 0.2\}$ (top, independence; bottom, the decorrelation channel of Section 2 at the Cohen–Maunsell [3] cortical anchor $r_{SC} \approx 0.2$). Columns: benefit/cost asymmetry $r \in \{0.3, 1, 3\}$. The vertical reference $V = 1/N = 0.25$ is the chance baseline; the $v = 1$ row is the value-blind reference where $VDA \equiv 0$ by construction. Filled contours are at $VDA \in \{0, 0.001, 0.005, 0.01, 0.025, 0.05, 0.1, 0.15, 0.2\}$ on a common scale. Substantial VDA is confined to a corner along low V and high v that flattens as r rises.

A quantitative threshold for spatial cueing designs. The contour band yields a concrete design prediction indexed on the validity threshold. Sweeping $v \in [1, 10]$ at fixed $r \in \{0.3, 1, 3\}$ and $\rho \in \{0, 0.2\}$ at four validity strata $V \in \{0.4, 0.6, 0.8, 0.95\}$, and recording the maximum VDA in each (V, r, ρ) stratum, gives Table 7. The table licenses three statements, each conditional on a quantitative validity threshold:

- **At $V \geq 0.95$,** peak VDA is at the grid floor for every (r, ρ) in the envelope: a high-validity design is predicted to carry negligible VDA at this strict threshold.
- **At $V \geq 0.80$,** peak VDA is at the grid floor under independence, but admits a small ρ -conditional signal at $\rho = 0.2$ (maximum 0.0032 at $r = 3$). “High validity” at this threshold therefore carries a ρ -conditional caveat: under cross-location correlation at the Cohen–Maunsell [3] anchor, a sub-percent signal is recoverable in the symmetric and benefit-dominant columns.
- **At $V \geq 0.60$,** the benefit is substantial: peak VDA reaches 0.1432 at $(r, \rho) = (0.3, 0)$ and 0.1639 at $(0.3, 0.2)$. Cost-dominant designs at $V \in [0.6, 0.8]$ are predicted to carry material VDA, concentrated at high cue value.

V	$r = 0.3$	$r = 1.0$	$r = 3.0$
0.40	0.1254 / 0.1197	0.1283 / 0.1394	0.0329 / 0.0391
0.60	0.1432 / 0.1639	0.0320 / 0.0461	0.0097 / 0.0120
0.80	0.0000 / 0.0000	0.0000 / 0.0023	0.0000 / 0.0032
0.95	0.0000 / 0.0000	0.0000 / 0.0000	0.0000 / 0.0000

Table 7: Peak VDA over the v -grid $[1, 10]$ at each (V, r) stratum, reported as the pair $\rho = 0 / \rho = 0.2$. Entries of 0.0000 are at the grid floor (no cell on the v -grid produced a non-trivial VDA at the allocation-grid resolution $\Delta\alpha = 0.005$). Bold entries flag the threshold transitions discussed in the text.

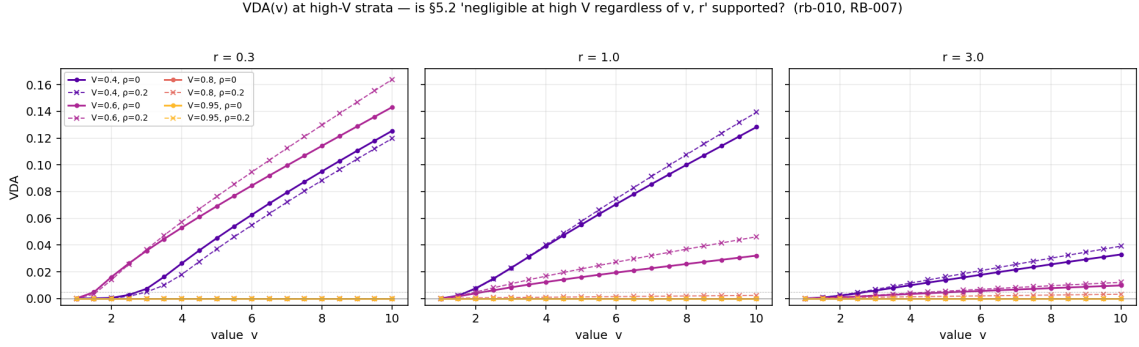


Figure 7: Peak VDA(v) at four validity strata $V \in \{0.4, 0.6, 0.8, 0.95\}$ (curve colour) for each $r \in \{0.3, 1, 3\}$ (panel column). Solid: $\rho = 0$; dashed: $\rho = 0.2$. The $V = 0.95$ curve is at the grid floor in every panel; the $V = 0.80$ curves split between a near-zero $\rho = 0$ line and a small $\rho = 0.2$ lift at $r \in \{1, 3\}$; the $V = 0.60$ curve at $r = 0.3$ reaches VDA = 0.16 at large v .

The model’s experimental-design guidance, anchored to Table 7 and the validity slices of Figure 7, is therefore:

Spatial cueing paradigms targeting a negligible-VDA regime should adopt $V \gtrsim 0.95$ unconditionally, or $V \gtrsim 0.8$ where cross-location correlation can be bounded below $r_{SC} \approx 0.2$. A threshold of $V \geq 0.75$ is too permissive: at $V \in [0.6, 0.8)$ the cost-dominant regime $r \in (0, 0.5)$ admits peak VDA up to ≈ 0.16 , with the largest values at high cue value v .

This guidance is stated at the headline configuration $(N, d'_{\max}, f_0, h) = (4, 2, 0.5, \sqrt{\cdot})$ under variant A; the variant B and conservation-family bands are taken up in the Discussion (Section 4). The validity slices of Figure 7 make the v -axis dependence explicit.

The decorrelation channel reshapes the band. The sign of $\partial\text{VDA}/\partial\rho$ is not uniform: across the design plane, decorrelation amplifies the benefit in some cells and suppresses it in others. Classifying each (V, v, r) cell by the sign of $\Delta\text{VDA} := \text{VDA}(\rho=0.2) - \text{VDA}(\rho=0)$ gives the signed contour map of Figure 8 and the per- r frequencies of Table 8. Two features stand out. First, the direction of the decorrelation effect organises along r : cost-dominant cells are suppression-dominated, while both the symmetric and benefit-dominant strata are amplification-dominated, so amplification is not a $r > 1$ phenomenon but the modal direction once r exceeds the $r \approx 0.4$ – 0.6 sign-flip locus identified in Section 3.2. Second, the largest single amplification in the entire sweep occurs at $(V, v, r) = (0.7, 10, 0.3)$: a high-validity cell that is essentially dormant under independence ($\text{VDA} = 0.0007$) lifts to $\text{VDA} = 0.0676$ at $\rho = 0.2$ — an $\approx 96\times$ amplification that carries a normally-negligible design point into the same magnitude as the low-validity corner. This dormant-cell amplification is a distinctive, falsifiable consequence of the decorrelation channel, taken up in the Discussion (Section 4).

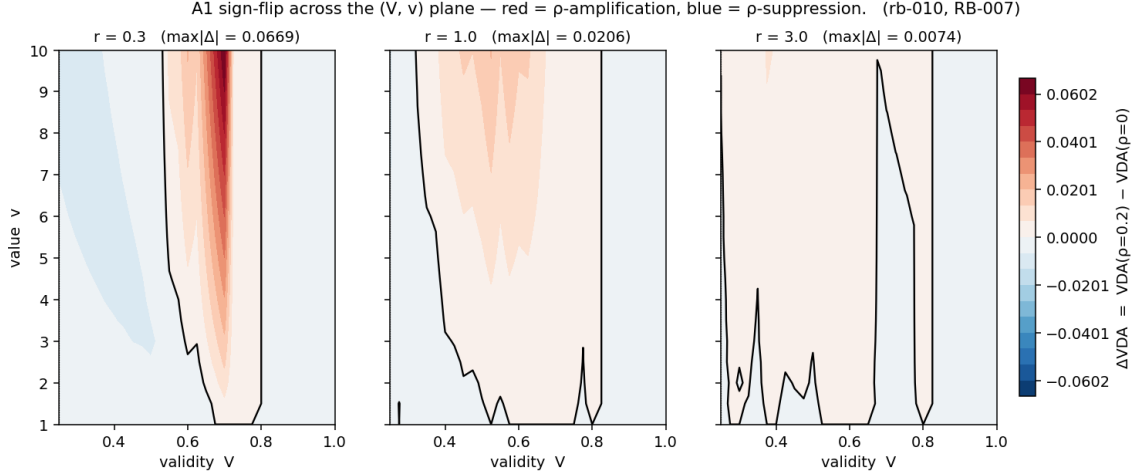


Figure 8: Signed contour map of $\Delta VDA := VDA(\rho=0.2) - VDA(\rho=0)$ over (V, v) for each $r \in \{0.3, 1, 3\}$. Red: ρ amplifies VDA; blue: ρ suppresses; white: $|\Delta VDA| < 10^{-8}$. At $r = 0.3$ (cost-dominant) suppression dominates at low V ; at $r \in \{1, 3\}$ amplification dominates across the bulk of the plane, with the strongest amplification near $(V, v) \approx (0.5, 10)$.

r	frac amp	frac supp	$\overline{\Delta VDA}$	max amp (V, v)	max supp (V, v)
0.3	27.2%	37.2%	+0.0012	+0.0669 (0.70, 10)	-0.0105 (0.25, 8.5)
1.0	52.8%	17.5%	+0.0023	+0.0206 (0.525, 10)	-0.0084 (0.25, 9)
3.0	54.0%	16.1%	+0.0009	+0.0074 (0.375, 10)	-0.0010 (0.25, 4.5)

Table 8: Cell-wise sign of $\partial VDA / \partial \rho$ across the (V, v) plane at each r stratum. Fractions are over the 589 cells per panel; the residual share is the zero-class $|\Delta VDA| < 10^{-8}$. Cost-dominant $r = 0.3$ cells are suppression-dominated; the symmetric and benefit-dominant strata are amplification-dominated.

Scope. These results are reported at the single configuration $(N, d'_{\max}, f_0, h) = (4, 2, 0.5, \sqrt{\cdot})$ under variant A and on the coarse correlation grid $\rho \in \{0, 0.2\}$. The variant B and conservation-family bands are developed in the Discussion (Section 4). The validity thresholds are bracketed by the V -grid step of 0.025; a finer grid would sharpen the “ $\gtrsim$ ” to a second-decimal number. As an internal consistency check, the $v = 1$ column is identically zero across every (V, r, ρ) cell: the value-blind baseline collapses the joint optimum onto the value-blind allocation, so $VDA = 0$ at $v = 1$ is a theorem of the model rather than a numerical coincidence.

3.4 Optimal allocation does not invert when the cue is predictive

A recurring question in value-cued attention is whether the normative observer ever does best to attend *away* from the high-value cued location — placing less than the uniform share $1/N$ on it. We find that under a predictive cue the answer is no: whenever the cued location carries at least its chance share of the targets, the optimal allocation α_c^* is at or above $1/N$ everywhere in the model’s primary operating range. The boundary that governs this fact is sharp and closed-form, and crossing it — making the cue *counter*-predictive — produces a clean inversion of the optimal policy that the model predicts quantitatively.

A value-weight inequality sets the no-inversion condition. The structural reason no inversion arises is a location-count asymmetry paired with a value-weight inequality. As $\alpha_c \rightarrow 1$ the single cued location reaches its per-location ceiling $d'_c = d'_{\max} f(1) = d'_{\max}$; as $\alpha_c \rightarrow 0$ the $N - 1$ uncued locations *share* the remaining budget at $(1 - \alpha_c)/(N - 1)$ each, so each attains only

$d'_u = d'_{\text{base}} + \beta(r)[d'_{\text{max}}f(1/(N-1)) - d'_{\text{base}}] \leq d'_{\text{max}}$ (strict for $N \geq 3$, where $f(1/(N-1)) < 1$). The per-channel reward weights are $w_c = Vv$ on the cued slot and $w_u = (1-V)/(N-1)$ on each uncued slot, and

$$w_c \geq w_u \iff V \geq \frac{1}{(N-1)v+1}. \quad (19)$$

For a predictive cue ($v \geq 1$) the right-hand side of (19) is bounded above by $1/N$, with equality only at $v = 1$. The condition $V \geq 1/N$ is therefore the universal (worst-case- v) sufficient condition for $w_c \geq w_u$, and the sharp form (19) carries strictly more slack as the cued value v grows. The right branch of the allocation dominates globally because the larger discriminability sits on the more-valuable channel.

A closed-form threshold organises the boundary. At $\alpha_c = 1/N$ the allocation derivative $d''(\alpha_c)$ has a kink, where the benefit/cost roles of β and γ swap which side of the linear $d'(\alpha_c)$ map is active. The one-sided left derivative of the expected reward at the uniform point takes the form

$$\left. \frac{\partial R_{P1}}{\partial \alpha_c} \right|_{1/N-} = \frac{2d'_{\text{max}}f'(1/N)}{r+1} \left[A_0(V, v, N, \text{CR}, \rho) - \frac{r}{N-1} B_0(V, v, N, \text{CR}, \rho) \right], \quad (20)$$

where A_0 and B_0 are the partial derivatives of the expected-reward functional with respect to d'_c and d'_u at $\alpha_c = 1/N$, evaluated at the jointly-optimised criteria (c_c^*, c_u^*) . Both are r -independent boundary partials — the only r -dependence in (20) sits in the prefactor and inside the bracket — so the bracket vanishes at a single closed-form threshold,

$$r_{\text{inv}}^*(V, v, N, \text{CR}, \rho) := (N-1) \frac{A_0(V, v, N, \text{CR}, \rho)}{B_0(V, v, N, \text{CR}, \rho)}. \quad (21)$$

For $r < r_{\text{inv}}^*$ the boundary left derivative is positive (the uniform point is locally preferred from the left branch); for $r > r_{\text{inv}}^*$ it is negative, so the left branch develops a local maximum at the allocation edge while the global maximum remains on the right branch at $\alpha_c^* > 1/N$. The boundary picture above r_{inv}^* is thus bimodal, with the value-weight inequality (19) deciding which branch is global. The threshold admits an exact anchor at the symmetric corner $(V, v) = (1/N, 1)$, where (19) holds with equality:

$$r_{\text{inv}}^*(V = 1/N, v = 1, N, \text{CR}, \rho) = 1 \quad \text{exactly}, \quad (22)$$

independent of N , the conservation rule CR, and the correlation ρ ; the identity is recovered to floating-point precision at every panel of Table 9. The full derivation of (20)–(22), including the corner identity from the joint first-order conditions, is given in the Supplementary material (Section A).

The threshold across the design grid. We evaluate (21) on the primary (V, v) grid ($V \in \{0.25, 0.30, \dots, 1.00\}$, $v \in \{1, 2, 3, 4, 5\}$) at $N = 4$ for both reward variants and $\rho \in \{0, 0.2\}$. Per panel we count cells with $r_{\text{inv}}^* \in [0.1, 10]$ (the asymmetry range over which the boundary derivative sign-flips) versus $r_{\text{inv}}^* > 10$ (no sign-flip in range; the local picture is monotone), and report the strict minimum and median (Table 9). Aggregated across all 210 $(V, v, \text{variant})$ cells at $\rho = 0$, 48.6% have $r_{\text{inv}}^* \in [0.1, 10]$; under the decorrelation channel at $\rho = 0.2$ the fraction rises to 51.9% and the median r_{inv}^* falls by $\sim 13\%$ (variant A) and $\sim 21\%$ (variant B), so decorrelation moves the boundary sign-flip to *smaller* r . The contour structure of $r_{\text{inv}}^*(V, v)$ at $\rho = 0$ is shown in Figure 9. The $r_{\text{inv}}^* = 1$ contour passes through the symmetric corner in both variants, consistent with (22), and the $r_{\text{inv}}^* = 10$ contour bounds the region in which the uniform point stays locally stable across the entire $r \in [0.1, 10]$ range.

variant	ρ	n	$r_{\text{inv}}^* \in [0.1, 10]$	$r_{\text{inv}}^* > 10$	$\min r_{\text{inv}}^*$	median r_{inv}^*
A	0.0	105	40 (38.1%)	65 (61.9%)	1.0000	17.30
A	0.2	105	42 (40.0%)	63 (60.0%)	1.0000	15.08
B	0.0	105	62 (59.0%)	43 (41.0%)	1.0000	7.64
B	0.2	105	68 (64.8%)	37 (35.2%)	1.0000	6.02

Table 9: Closed-form inversion threshold r_{inv}^* from (21) on the primary (V, v) grid at $N = 4$, by reward variant and correlation ρ . The $r_{\text{inv}}^* = 1.0000$ minimum in every panel is the symmetric-corner identity (22). Variant B admits the local sign-flip across roughly twice as many cells as variant A; the decorrelation channel ($\rho = 0.2$) shifts the sign-flip to smaller r , lowering the median threshold in both variants.

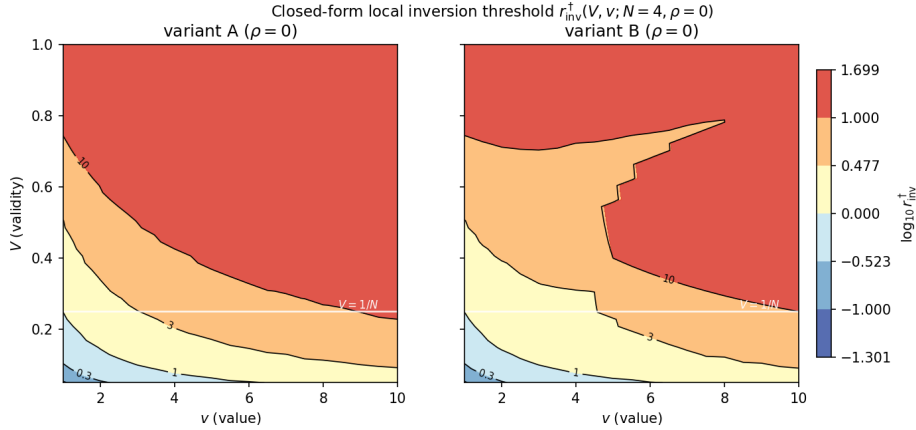


Figure 9: Contour map of $\log_{10} r_{\text{inv}}^*(V, v)$ at $N = 4$, $\rho = 0$, for reward variant A (left) and variant B (right), evaluated from (21)–(22). Black contours at $r_{\text{inv}}^* \in \{0.1, 0.3, 1, 3, 10\}$. The $r_{\text{inv}}^* = 1$ contour passes through the symmetric corner $(V, v) = (0.25, 1)$ in both panels (the identity (22)); the $r_{\text{inv}}^* = 10$ contour bounds the region with no boundary sign-flip inside $r \in [0.1, 10]$. Variant B admits the sign-flip across more cells than variant A (59.0% vs. 38.1%; Table 9).

No global inversion across the predictive-cue sweep. The local sign-flip of (20) does not by itself produce a global inversion: even where $r > r_{\text{inv}}^*$ and the bracket is negative, the left-branch local maximum sits at the allocation edge while the right-branch global maximum sits at $\alpha_c^* > 1/N$, by the location-count argument above. We confirm this directly by sweeping $R(\alpha_c)$ on the full allocation grid $[0.02, 1.0]$ at step 0.005 for the most adversarial primary-sweep cells (those with the smallest r_{inv}^* at $r = 10$, most likely to favour the left branch), at both $\rho \in \{0, 0.2\}$. Across all 12 probes the result is uniform: $\alpha_c^* \geq 1/N$, and the right-branch global maximum strictly dominates (Table 10). The no-inversion property holds at $\rho = 0$ and is unchanged under the decorrelation channel at $\rho = 0.2$. The boundary geometry is illustrated in Figure 10, which plots $R(\alpha_c)$ for a family of asymmetry ratios. The kink at $\alpha_c = 1/N$ becomes a local minimum once $r > r_{\text{inv}}^*$, with local maxima on both branches; which branch wins globally is set by the value-weight inequality (19). The panel is drawn at a counter-predictive cell $(V, v) = (0.15, 1)$, where (19) fails, so the reader sees both regimes of the boundary at once — the subject of the next paragraph.

Counter-predictive cues invert the optimal policy. The model makes a clean prediction once the cue is counter-predictive. At $V < 1/N$ (sharply, $V < 1/[(N-1)v + 1]$) the value-weight inequality (19) flips, $w_c < w_u$, and the location-count argument runs in reverse: the larger discriminability would sit on the less-valuable channel, so the global optimum drops to

V	v	variant	ρ	r_{inv}^*	α_c^*	R_{global}	R_{left}
0.250	1	A	0.0	1.0000	0.955	0.65936	0.64564
0.250	1	A	0.2	1.0000	0.950	0.66809	0.65550
0.250	1	B	0.0	1.0000	0.955	0.65936	0.64564
0.250	1	B	0.2	1.0000	0.950	0.66809	0.65550
0.2875	1	A	0.0	1.2483	0.970	0.66541	0.64462
0.325	1	A	0.2	~ 1.6	0.975	0.67997	—

Table 10: Six representative full-allocation probes from the 12-probe sweep at $r = 10$ at the most adversarial predictive-cue cells. In every probe $\alpha_c^* \geq 1/N = 0.25$, with the right-branch global maximum at $\alpha_c^* \in [0.95, 0.98]$ strictly dominating the left-branch local maximum (at the grid edge $\alpha_c = 0.02$). Across all 12 probes the number of global inversions is **0**, at both $\rho = 0$ and $\rho = 0.2$.

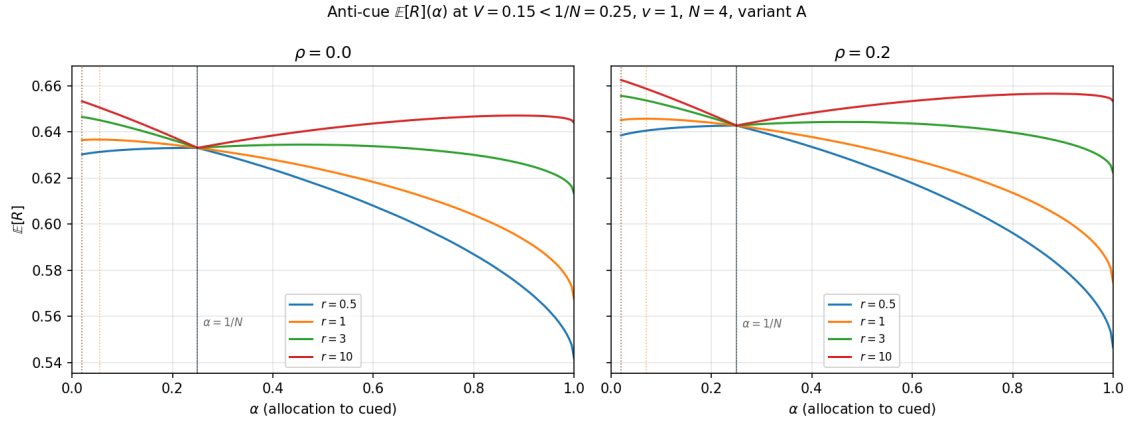


Figure 10: Expected reward $R(\alpha_c)$ on the full allocation grid at the counter-predictive cell $(V, v, N) = (0.15, 1, 4)$ for $r \in \{0.5, 1, 3, 10\}$, at $\rho = 0$ (left) and $\rho = 0.2$ (right). The kink at $\alpha_c = 1/N = 0.25$ is the benefit/cost swap of the allocation map; for $r > r_{\text{inv}}^*$ it is a local minimum, with both branches admitting local maxima. Because $V < 1/N$ here, the value-weight inequality (19) fails and the left-branch maximum (at the grid edge $\alpha_c = 0.02$) dominates globally for $r \geq 1$ — the counter-predictive inversion. The decorrelation channel at $\rho = 0.2$ leaves the bimodality and the inversion onset unchanged.

$\alpha_c^* < 1/N$ and the normative observer attends *away* from the low-validity location toward the population of alternatives. We probe a counter-predictive sub-grid $V \in \{0.05, 0.10, 0.15, 0.20\}$ (all below $1/N = 0.25$), $v \in \{1, 3, 5\}$, $r \in \{0.1, 0.5, 1, 3, 5, 10\}$, variant A, at $\rho \in \{0, 0.2\}$. The global-inversion incidence is $26/72 = 36.1\%$ at $\rho = 0$ and $25/72 = 34.7\%$ at $\rho = 0.2$, and it stratifies cleanly along the sharp boundary (19) (Table 11). Figure 11 maps $\alpha_c^*(V, r)$ at fixed $v = 5$. The inversion region lies entirely below $V = 0.05$, matching the sharp boundary $V < 1/16 \approx 0.0625$, and there are *zero* inversions in the predictive-cue regime $V \geq 1/N$ at either correlation. The no-inversion property is thus a conditional theorem — it holds whenever the cue is predictive — and the counter-predictive inversion is a falsifiable prediction the model makes outside that regime.

Robustness and decorrelation sensitivity. The no-inversion result is robust across the model’s parameters. The empirical sweep finds $\alpha_c^* \geq 1/N$ at every cell of the predictive-cue range under both reward variants, and the decorrelation channel does not abolish the picture: at $\rho = 0.2$ the median threshold r_{inv}^* shifts by 13–21%, the predictive-cue sweep still returns 0 inversions, and the counter-predictive incidence moves by at most one cell (25 vs. 26 inversions).

Inversion incidence at $\rho = 0$ (values $\rightarrow$ inversions / total)						
by r	$r=0.1$: 1/12	$r=0.5$: 3/12	$r=1$: 6/12	$r=3$: 6/12	$r=5$: 6/12	$r=10$: 4/12
by v	$v=1$: 18/24	$v=3$: 5/24	$v=5$: 3/24			
by V	$V=0.05$: 14/18	$V=0.10$: 5/18	$V=0.15$: 4/18	$V=0.20$: 3/18		

Table 11: Counter-predictive inversion incidence ($\alpha_c^* < 1/N$) at $N = 4$, variant A, $\rho = 0$, stratified by r , v , and V (total $26/72 = 36.1\%$). Incidence rises from 8.3% at $r = 0.1$ to 50% at $r \in \{1, 3, 5\}$ and tapers to 33.3% at $r = 10$, where the strong-asymmetry regime begins to re-favour the right branch. The value dependence is the key qualitative signature predicted by (19): inversion is dense at $v = 1$ (75%) but rare at $v = 5$ (12.5%), because at $v = 5$ the inversion boundary shrinks to $V < 1/[(N-1)v+1] = 1/16 = 0.0625$, confining inversion to the $V = 0.05$ row. Under the decorrelation channel ($\rho = 0.2$) total incidence is 34.7%, within one cell of $\rho = 0$ across every stratifier.

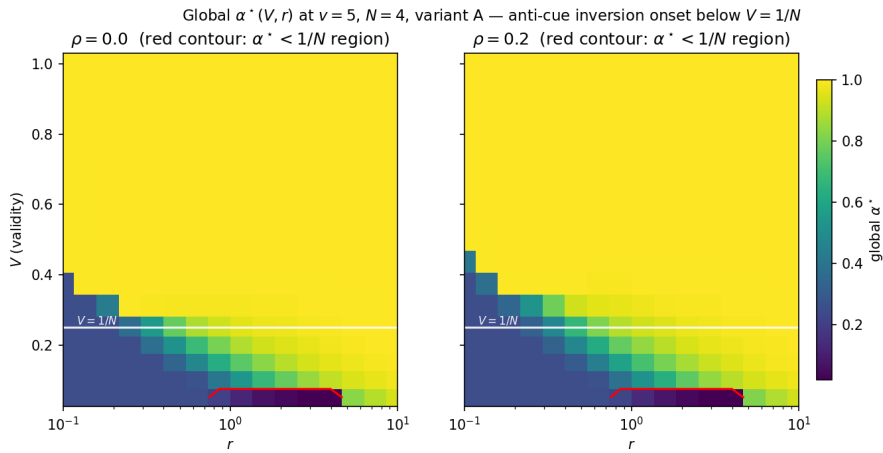


Figure 11: Optimal allocation $\alpha_c^*(V, r)$ at $(N, d'_{\max}, f_0, h, v) = (4, 2, 0.5, \sqrt{\cdot}, 5)$, variant A, for $\rho = 0$ (left) and $\rho = 0.2$ (right); heat scale $\alpha_c^* \in [0.02, 1.0]$. The white horizontal line marks the validity threshold $V = 1/N = 0.25$; the red contour bounds the inversion region $\alpha_c^* < 1/N$. The inversion region is confined to $V \leq 0.05$ (consistent with the sharp boundary $V < 1/16 \approx 0.0625$ from (19)); inversion incidence is $6/272 = 2.21\%$ in both panels, with *zero* inversions in the predictive-cue regime $V \geq 1/N$.

Decorrelation and the inversion lever are therefore independent: ρ moves *where* on the r -axis the boundary derivative sign-flips, but it does not flip the value-weight inequality (19) that decides which branch is globally optimal, so the conditional theorem and its counter-predictive complement hold at both correlation values.

Relation to the behavioural record. The conditional picture aligns with the behavioural literature on spatial value and statistical learning. When a location holds a target with below-chance probability, observers learn to suppress it: capture and target-selection efficiency at the location fall [12], fewer saccades land there and latencies to targets there rise [13], and selection is reallocated toward the alternatives [14]. In the model's coordinates a below-chance location has validity $V < 1/N$ — exactly the regime in which (19) predicts inversion — so the behavioural pattern is a prediction match: the normative observer should allocate below uniform there, and observers do. Conversely, in the predictive-cue regime $V \geq 1/N$ the model says the observer should never go below uniform, and the value-driven-capture literature reports the opposite mistake, observers pulled *toward* high-value locations even when maladaptive [4, 5], consistent with the no-inversion direction. The chance-validity case $V = 1/N$ [2] sits at the boundary,

where the model collapses to a tied optimum and no reallocation is predicted.

Scope. The counter-predictive sweep is reported for variant A; the lower median r_{inv}^* of variant B (Table 9) indicates a somewhat higher inversion incidence there, mapped in the Discussion (Section 4). The validity grid is at step 0.05, so the inversion-onset validity between $V = 0.20$ and the boundary $V = 0.25$ at $v = 1$ is bracketed to ± 0.05 ; a finer grid would localise it. As a consistency check, the symmetric corner $(V, v) = (1/N, 1)$ returns $r_{\text{inv}}^* = 1$ to floating-point precision across all variants and correlations (22), and at $v = 1$ the model is value-blind so the predictive and counter-predictive regimes meet at a tied optimum.

4 Discussion

We set out to ask when value-directed attention matters: under what conditions does re-allocating perceptual resources toward a high-value location gain meaningfully over an observer that has already optimised its decision criteria and its validity-driven attention? The model returns a layered answer. Across a broad parameter space, criterion adjustment is typically the dominant lever for encoding value, capturing a median of roughly three-quarters of the reward gain (Section 3.1). The advantage of value-directed attention, where it exists, is non-monotonic in the benefit/cost ratio r and is governed by a closed-form lower edge $r^\dagger(v)$ (Section 3.2); it is concentrated in a graded corner of the design space rather than distributed across it (Section 3.3); and the direction of optimal allocation never inverts when the cue is predictive, a conditional theorem holding for $V \geq 1/N$ (Section 3.4). Together these results describe a task class in which the neural machinery for value-driven re-allocation is, over much of the natural operating range, redundant with a simpler criterion-based strategy — and they delimit precisely where it is not.

Why criterion adjustment is typically dominant

Criterion adjustment and value-directed attention exploit the same cued information by different routes, and the routes differ in price. Shifting the decision threshold at a high-value location requires no perceptual reallocation: a more liberal criterion there trades false alarms for hits while leaving sensitivity at every other location untouched. Spatial re-allocation, by contrast, pays a genuine perceptual cost — gain acquired at the cued location is taken from the others. When the cue is predictive, validity-driven attention has already concentrated processing where it is most likely to be needed, and the residual value information is most economically booked into the decision thresholds rather than into further perceptual gain. This is why, across the sweep, the criterion fraction CF concentrates well above one-half (median 0.7552 for the value-coupled reward variant and 0.7682 for the equal-reward variant, both at the additive conservation rule), with most of the distribution lying in $[0.30, 1.00]$ (Section 3.1). The lever does not dominate everywhere: in the benefit-dominant, low-validity corner — where attentional enhancement is efficient and the cue is weakly informative — the criterion fraction falls and value-directed attention contributes materially. The summary is therefore a central tendency with a substantial tail across the parameter space.

The benefit/cost asymmetry and its biological reading

The benefit/cost ratio r is the model’s compact summary of a dissociation observed in cortex: attentional enhancement at attended locations and suppression at unattended locations are separately modulable, and the normalisation framework that captures single-unit gain effects treats them as distinct operations rather than two ends of one knob [8, 10, 9, 15]. A large r describes an enhancement-dominant regime in which attending a location buys more sensitivity

than is lost elsewhere; a small r describes a suppression- or cost-dominant regime in which re-allocation is expensive. The non-monotonicity of the value-directed advantage in r (Section 3.2) then has a clean reading. When enhancement is cheap (large r), a value-blind observer already attends the cued location and value adds little; when enhancement is very expensive (small r), no observer should re-allocate and again value adds little. The advantage peaks in between, in a cost-dominant window where value-blind attention is too expensive to deploy but the value differential makes targeted re-allocation worthwhile — exactly the band whose lower edge the closed-form threshold $r^\dagger(v)$ marks (16). The model thus locates value-directed attention not where attention is easiest but where value is the deciding factor, which is a more demanding and more diagnostic regime than the one many designs target.

Guidance for experimental design

These results carry directly into how one should design an experiment to observe value-directed attention. Because the advantage is concentrated in a graded corner — low cue validity, high value contrast, cost-dominant asymmetry, and low baseline sensitivity (Section 3.3) — a design placed outside that corner will tend to produce null value-directed effects even when the underlying mechanism is intact, simply because optimal criterion adjustment already captures nearly all of the available reward. Many standard spatial-cueing paradigms, which use highly valid cues and moderate value manipulations, operate in this dormant regime. The model makes the boundary quantitative: at cue validity $V \geq 0.95$ the predicted advantage sits at the grid floor for every asymmetry and correlation we examined; the dormancy survives down to $V \geq 0.80$ when cross-location correlation is negligible but admits a small correlation-conditional signal otherwise; and a $V \geq 0.75$ criterion is too permissive, because cost-dominant designs in $V \in [0.60, 0.80)$ admit a peak advantage near 0.16 (Section 3.3). An experiment aiming to isolate value-driven attention from criterion-based strategy should therefore move deliberately into the cost-dominant, low-to-moderate-validity, high-contrast corner rather than maximising cue validity.

New predictions

The model makes three predictions that go beyond restating where the advantage lives. First, although optimal allocation never inverts under a predictive cue, it *can* invert under a counter-predictive “anti-cue”: when validity falls below chance, $V < 1/N$, re-allocating attention *away* from the nominally cued location becomes optimal in a substantial fraction of cells (Section 3.4). The inversion is sharply bounded in value contrast — it is confined below the closed-form boundary $V < 1/[(N-1)v+1]$ — and is absent throughout the predictive regime, which makes it a clean, falsifiable signature: an observer who is genuinely re-allocating perceptual resources, rather than only shifting criteria, should show inverted allocation under an anti-predictive cue and never under a predictive one.

Second, the cross-location correlation ρ is an active lever, not a nuisance term, and its effect on the value-directed advantage changes sign with the benefit/cost ratio. In cost-dominant designs (r small) raising ρ suppresses the advantage, whereas in balanced-to-enhancement-dominant designs ($r \gtrsim 1$) it amplifies it (Section 3.3). The amplification can be dramatic at design points that look dormant under independent noise: at one cost-dominant high-contrast cell the advantage rises from a negligible 0.0007 at $\rho = 0$ to 0.0676 at $\rho = 0.2$, roughly a hundredfold increase driven entirely by admitting the empirically typical level of inter-neuronal correlation [3, 6, 7]. The prediction is that whether a given design reveals value-directed attention can depend on the correlation structure of the neural code, so that a paradigm that is null under one correlation regime may not be under another.

Third, the position of the criterion/attention split is sensitive to the form of the benefit/cost conservation rule. Replacing the additive rule ($\beta + \gamma = 2$) with the multiplicative endpoint

($\beta \cdot \gamma = 1$) leaves the central tendency of CF almost unchanged — the median moves by less than 0.005 — but it deepens the tail: the fraction of cells with $CF < 0.5$ roughly doubles, and the per-cell criterion fraction never rises under the multiplicative rule (Section 3.1). The dominance of criterion adjustment is thus robust as a central claim but rule-dependent in its tail, and the size of the attention-favourable tail is itself an observable consequence of how enhancement and suppression trade off — a structural prediction about the resource constraint, not a free parameter. The equal-reward convention sharpens the same point: its deeper CF tail and its lower median inversion threshold (Table 9) indicate a somewhat larger attention-favourable region and a somewhat higher anti-cue inversion incidence than the value-coupled reward variant, so the reward structure of a task is itself a design variable that shifts the regime boundaries.

Scope and limitations

Several boundaries delimit these statements. The benefit/cost conservation constraint relating the benefit and cost weights (Section 2, Equation (4)) is one member of a one-parameter family spanning the additive ($\beta + \gamma = 2$) and multiplicative ($\beta \cdot \gamma = 1$) endpoints; the headline numbers are reported under the additive rule with a band over the family, and which member corresponds to a particular biological resource constraint is not fixed by the model. The single asymmetry ratio r summarises a homogeneous population; lifting it to a per-location vector and the uncued allocation to the full N -dimensional simplex leaves the optima essentially intact under modest heterogeneity, but the closed-form dependence of the optimum on the spread of r across locations is not derived here. The decision rule treats per-location decision noise as fixed; an attention-coupled noise channel $\sigma_i(\alpha)$ would, if it were operative, enter alongside decorrelation as a further axis of the decomposition and would shift the interpretation of the criterion fraction in the cost-dominant regime, and we leave it as a natural extension rather than including it. The transfer function relating attention to sensitivity is taken from a family of monotonic forms; the mechanism statements (Definition 1, Equation (16)) are stated for a generic transfer function and do not depend on the particular family, though the empirical bands do. The correlation lever assumes a single equicorrelated structure; distance- or feature-dependent covariances break the one-factor reduction and are out of scope. Finally, the model is normative and stationary: it describes the optimal policy for an observer at a fixed parameter setting, not how that policy is learned, and it makes no claim about the neural implementation of any of the three levers beyond the correspondence between the asymmetry ratio and the enhancement/suppression operations of cortical gain control. Its contribution is a quantitative map of when value-directed attention is the deciding mechanism and when an optimally-set criterion suffices — a map that a downstream empirical or computational program can use to place its experiments where the mechanism it cares about is actually at stake.

5 Methods

5.1 Task and decision model

All results derive from the signal-detection model of Section 2. On each trial a change is present with probability 0.5 and, when present, occurs at the cued location with probability V and at each of the $N - 1$ uncued locations with probability $(1 - V)/(N - 1)$. Each location carries an equal-variance Gaussian decision variable, and the observer applies an independent detection criterion c_i at every location, yielding the per-location hit and false-alarm rates of Equation (1). Detecting a change at the cued location earns reward $v \geq 1$; detecting a change at any uncued location earns reward 1; a correct rejection earns CR. Expected reward on a trial is Equation (7). Unless stated otherwise, all reported sweeps fix the perceptual cell at $N = 4$, maximal sensitivity $d'_{\max} = 2$, baseline fraction $f_0 = 0.5$, and the square-root transfer function

$h(a) = \sqrt{a}$ in Equation (2); the remaining four functional forms of Equation (3) are used for the transfer-function robustness checks.

5.2 Benefit, cost, and reward variants

The asymmetry between attentional enhancement and suppression is set by the ratio $r = \beta/\gamma$, with the benefit and cost weights fixed by the additive conservation rule of Equation (4), $\beta + \gamma = 2$. Sensitivity to the form of this conservation constraint is examined as a one-parameter power-mean family that contains the additive rule as a special case; the family and the resulting band on the criterion fraction CF are given in the Supplementary material (Section A). Two reward variants are evaluated throughout to separate task-design effects from incentive structure: *variant A*, in which the correct-rejection reward is value-coupled, $CR = Vv + (1 - V)$; and *variant B*, in which it is fixed at $CR = 1$. Variant A is the primary cell for the figures; variant B is reported alongside it wherever the two diverge.

5.3 Correlated-noise channel

Cross-location noise is admitted through the equicorrelation parameter $\rho \in [0, 1)$ of Equation (9). The only cross-location term in the expected reward is the no-false-alarm probability $P_{\text{no-fa}}$, and equicorrelation collapses its N -dimensional orthant integral to the one-dimensional integral of Equation (10). We evaluate that integral by 64-node Gauss-Hermite quadrature. The quadrature is converged: across the operating band $\rho \in \{0, 0.05, 0.1, 0.2, 0.3, 0.4\}$ the 64-node estimate agrees with a 128-node reference to within 10^{-15} , several orders of magnitude below the precision of any reported quantity. All headline results use $\rho \in \{0, 0.2\}$, bracketing the independent-noise limit and the $\rho \approx 0.2$ spike-count correlation reported in macaque V4 during peripheral-cue change detection [3].

5.4 Policy optimisation

The three-lever decomposition (Definition 1) is computed from the four nested policies R_{P1} through R_{P4} defined in Section 2.5. Each policy is evaluated by exhaustive grid optimisation, which is robust to the non-convexity of the expected-reward surface in the attention share:

- **Criteria.** At any sensitivity pair (d'_c, d'_u) , the per-location criteria are optimised over a grid of 121 values, $c_i \in [-3, 3]$ in steps of $\Delta c = 0.05$, and the pair (c_c, c_u) maximising Equation (7) is selected. The floor policy R_{P4} instead fixes $c = 0$.
- **Attention.** The cued share α_c is optimised over a grid spanning the admissible range, with the uncued share following from $\alpha_c + (N - 1)\alpha_u = 1$. For the joint optimum R_{P1} and the value-blind allocation R_{P2} , α_c ranges over $[1/N, 1]$; the distributional and anti-cue sweeps extend the grid to $[0.02, 1]$ so that allocations below uniform ($\alpha_c < 1/N$) are admissible and inversion can be detected when it occurs. The uniform-attention policies R_{P3}, R_{P4} fix $\alpha_c = 1/N$.

The value-directed-attention benefit is $VDA = R_{P1} - R_{P2}$ (Equation (14)) and the criterion fraction is CF (Equation (15)). The value-blind allocation that defines R_{P2} depends only on $(r, V, N, d'_{\text{max}}, f_0, h, \text{variant}, \rho)$ and is therefore computed once per such configuration and reused across the value grid.

5.5 Parameter sweeps

Four sweeps support the four findings; all fix the perceptual cell ($N = 4$, $d'_{\text{max}} = 2$, $f_0 = 0.5$, $h = \sqrt{\cdot}$) and run at $\rho \in \{0, 0.2\}$.

1. **Distributional sweep (criterion fraction).** A primary grid of 22 benefit/cost ratios r (log-spaced over $[0.1, 10]$, with $r = 1$ pinned) $\times$ 21 validities V (uniformly spaced over $[1/N, 1]$) $\times$ 5 values $v \in \{1, 2, 3, 4, 5\}$ $\times$ 2 reward variants. Of the 4,620 nominal cells, 4,410 (2,205 per variant) yield valid non-degenerate optima and enter the CF distribution.
2. **Value-directed-attention curves.** The benefit VDA as a function of r on a high-resolution grid of 84 ratios (83 log-spaced, with the escape and peak neighbourhoods pinned), for a value family $v \in \{2, 3, 5, 8, 10\}$, used to locate the peak ratio r^* and compare it against the closed-form escape threshold $r^\dagger(v)$ of Equation (16).
3. **Iso-VDA maps.** A $31 \times 19 \times 3 \times 2 = 3,534$ cell sweep over $V \in [0.25, 1]$ (31 points), $v \in [1, 10]$ (19 points), $r \in \{0.3, 1, 3\}$, and $\rho \in \{0, 0.2\}$, used to draw the iso-VDA contour band over the (V, v) experimental-design plane and to evaluate the high-validity design recommendation.
4. **Inversion analysis.** The closed-form inversion threshold $r_{\text{inv}}^*(V, v, N, \text{CR}, \rho)$ of Equation (21) is evaluated on the primary (V, v) grid ($21 \times 5 \times 2$ variants $\times$ 2 correlations = 420 cells). The optimal allocation is then verified directly by sweeping R_{P1} over the extended attention grid (197 points on $[0.02, 1]$) at the most adversarial high-ratio cells, and the counter-predictive regime is probed on a dedicated grid of validities below chance, $V \in \{0.05, 0.10, 0.15, 0.20\} \times v \in \{1, 3, 5\} \times r \in \{0.1, 0.5, 1, 3, 5, 10\} \times \rho \in \{0, 0.2\}$. The allocation map $\alpha_c^*(V, r)$ at $v = 5$ is computed on a $17 \times 16 \times 2 = 544$ cell grid.

Robustness checks vary the transfer function over the four forms of Equation (3), the location count N , and the conservation form (Section A).

5.6 Validation

Two internal consistency checks anchor the correlated-noise implementation. First, the *recovery contract* (Equation (11)): at $\rho = 0$ the integrand of Equation (10) is constant in the latent factor, and the quadrature reduces to the closed-form product Equation (8). Evaluated numerically, $P_{\text{no-fa}}$, the expected reward, VDA, and CF at $\rho = 0$ coincide with the independent-noise computation to floating-point precision (maximum absolute difference below 10^{-6} across the full distributional grid), so every reported quantity contains the independent-noise case as its exact $\rho = 0$ limit. Second, the closed-form thresholds are cross-checked against the grid optima they predict: the escape threshold $r^\dagger(v)$ orders correctly against the empirically located peak ratio r^* across the value family (Section 3), and the inversion threshold satisfies the exact symmetric-corner identity $r_{\text{inv}}^*(1/N, 1, N, \text{CR}, \rho) = 1$ (Equation (22)) independent of variant and ρ . By the Slepian inequality [11], $P_{\text{no-fa}}(\rho)$ is non-decreasing in ρ at fixed criteria and sensitivities, providing a sign check on the correlated channel.

5.7 Reproducibility

All computations are deterministic. No Monte-Carlo sampling is used: the expected-reward expression is evaluated in closed form up to the one-dimensional Gauss–Hermite quadrature, and every optimisation is an exhaustive search over fixed grids. The parameter grids, quadrature nodes, and standard-normal evaluations are fixed across runs, so each sweep reproduces identical numerical output on re-execution. The full distributional sweep completes in under two minutes on a contemporary laptop. Computations were carried out in Python using standard numerical and special-function libraries.

A Supplementary derivations and extensions

This material derives the results used in the main text and develops the model’s scope. Section A.1 derives the one-dimensional integral for the no-false-alarm probability under correlated decision noise and establishes its monotonicity in the correlation. Section A.2 derives the closed-form escape threshold for value-directed attention and extends it across the correlation family. Section A.3 records the exact recovery of the symmetric model at unit benefit/cost ratio. Section A.4 develops the one-parameter conservation family that generalises the benefit/cost weighting and establishes the invariances on which the headline results rest. Throughout, φ is the standard normal density, Φ its cumulative distribution function, and $b_i := c_i + d'_i/2$ the z -score the no-change decision variable must stay below at location i .

A.1 The correlated no-false-alarm probability and its monotonicity

The only cross-location term in the expected reward (7) is the probability $P_{\text{no-fa}}$ that no location produces a false alarm on a no-change trial (Section 2.4). Under independent per-location decision variables this is the product (8); here we derive its exact form when the decision variables share an equicorrelation ρ and show that admitting positive correlation can only raise it.

The exact one-dimensional reduction. Let the no-change decision variables be jointly Gaussian with standardised marginals and equicorrelation, as in (9). The one-factor representation

$$X_i = \mu_i + \sqrt{\rho} Z + \sqrt{1-\rho} \varepsilon_i, \quad Z, \varepsilon_1, \dots, \varepsilon_N \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1), \quad (23)$$

with $\mu_i = -d'_i/2$, reproduces $\text{Var}(X_i) = 1$ and $\text{Cov}(X_i, X_j) = \rho$ for $i \neq j$, and makes the X_i conditionally independent given the shared latent Z . Conditional on $Z = z$, the per-location no-false-alarm event $\{X_i \leq c_i\}$ has probability

$$\Pr(X_i \leq c_i \mid Z = z) = \Phi\left(\frac{c_i - \mu_i - \sqrt{\rho} z}{\sqrt{1-\rho}}\right) = \Phi\left(\frac{b_i - \sqrt{\rho} z}{\sqrt{1-\rho}}\right). \quad (24)$$

Multiplying across the conditionally independent locations and integrating Z out against φ gives the exact one-dimensional integral (10). Equation (10) holds for any N and any $\rho \in [0, 1]$: equicorrelation collapses the N -dimensional orthant integral to a single dimension, with no multivariate-normal cumulative distribution function required. At $\rho = 0$ the integrand of (10) is constant in z and the integral reduces to the independent product (11). The model evaluates (10) by 64-node Gauss–Hermite quadrature; with the substitution $z = \sqrt{2}t$ the integrand carries weight e^{-t^2} , and the 64-node rule agrees with a 128-node reference to better than 10^{-15} across the operating band $\rho \in \{0, 0.05, 0.1, 0.2, 0.3, 0.4\}$.

Monotonicity in the correlation.

Proposition 1 (Orthant monotonicity). *For jointly Gaussian decision variables with standardised marginals and equicorrelation $\rho \in [0, 1]$, the no-false-alarm probability $P_{\text{no-fa}}(\rho)$ of (10) is non-decreasing in ρ at every fixed threshold pair (b_c, b_u) , with strict increase for $\rho > 0$ at finite thresholds:*

$$P_{\text{no-fa}}(\rho) \geq P_{\text{no-fa}}(0) = \Phi(b_c) \Phi(b_u)^{N-1} \quad \forall \rho \in [0, 1]. \quad (25)$$

Proof. Equation (25) is the equicorrelated case of Slepian’s monotonicity inequality for Gaussian orthant probabilities, which states that $\Pr(\bigcap_i \{X_i \leq c_i\})$ is non-decreasing in each off-diagonal correlation entry at fixed thresholds [11]; applying it to every pair correlation simultaneously gives monotonicity in the common value ρ . A modern treatment is given in [16]. $\square$

Positive correlation relaxes the multiple-comparisons pressure that, in the independent model, drives the N -fold no-false-alarm product toward zero whenever any single $\Phi(b_i)$ falls below one. The independent corner $\rho = 0$ is therefore the stiffest-penalty boundary of the family: it maximises the aggregate cost of holding N false-alarm rates below threshold at once. Because the change-trial bracket of (7) involves only marginal hit rates and is independent of ρ , and the no-change bracket scales linearly in $P_{\text{no-fa}}(\rho)$ with the non-negative coefficient $\frac{1}{2}\text{CR}$, Proposition 1 gives a pointwise reward bound that survives the supremum over criteria.

Corollary 2 (Per-policy reward monotonicity). *At any fixed allocation α , the criterion-optimised expected reward $R^*(\alpha; \rho) := \sup_{c_c, c_u} \mathbb{E}[R](\alpha, c_c, c_u; \rho)$ is non-decreasing in ρ . Consequently each of the nested policies $P1$ – $P4$ of Section 2.5 has its optimal reward non-decreasing in ρ .*

Corollary 2 bounds the reward *levels* but not their *difference*: the value-directed-attention benefit $\text{VDA}(\rho) = R_{P1}(\rho) - R_{P2}(\rho)$ is the gap between two quantities that both rise with ρ , and such a gap can move in either direction. The sign of $\partial \text{VDA} / \partial \rho$ is therefore a property of the model to be determined, not a corollary of monotonicity. Two mechanisms compete. Raising ρ inflates $P_{\text{no-fa}}(\rho)$ most at liberal criteria, flattening the no-change bracket across (c_c, c_u) and *devaluing* the criterion lever; this is the channel that drives the criterion fraction $\text{CF}(\rho)$ downward (Section 3.1). At the same time, when the optimal allocation concentrates attention on the cued location the uncued sensitivity falls and the uncued false-alarm rate rises, compounded across $N - 1$ locations through the no-false-alarm product; Proposition 1 shows this aggregate penalty is *relaxed* under correlation, making concentration cheaper and *amplifying* VDA. The criterion-devaluation channel dominates in the cost-dominant regime (small r) and the concentration-relaxation channel dominates in the benefit-dominant regime (large r), so the sign of $\partial \text{VDA} / \partial \rho$ flips with r — the model property reported across the parameter plane in Sections 3.2 and 3.3.

A.2 Closed-form escape threshold for value-directed attention

Here we derive the escape threshold $r^\dagger(v)$ of (16)–(18) and extend it across the correlation family. The threshold is the value of the benefit/cost ratio below which the value-aware optimum is still pinned at uniform allocation $\alpha = 1/N$, and above which the optimiser begins to move sensitivity toward the cued location.

Setup at the boundary. At the uniform allocation $\alpha = 1/N$ the asymmetric sensitivity rule of Section 2.3 collapses: $d'_c = d'_u = d'_{\text{base}} := d'_{\text{max}} f(1/N)$, independent of (r, ρ) , because the gain bracket $d'_{\text{max}} f(1/N) - d'_{\text{base}}$ vanishes before being weighted by $\beta(r)$ or $\gamma(r)$. The criterion optimum at this allocation is therefore the joint maximiser of the expected reward over a fixed- d' landscape that depends on ρ only through the correlated no-false-alarm term:

$$(c_c^*(\rho), c_u^*(\rho)) := \arg \max_{(c_c, c_u)} R(d'_{\text{base}}, d'_{\text{base}}, c_c, c_u; V, v, N, \text{CR}, \rho). \quad (26)$$

At the headline cell $(N, V, v) = (4, 0.5, 5)$ with $\text{CR} = Vv + (1 - V) = 3.0$, the optimal pair $(c_c^*, c_u^*) = (0.10, 1.75)$ at $\rho = 0$ shifts to $(0.05, 1.80)$ at $\rho = 0.2$: the cued criterion becomes more liberal (a cued false alarm is partially explained away by correlated false alarms at the other locations) and the uncued criterion more conservative (an uncued false alarm now also degrades the cued no-false-alarm joint event).

The correlation-aware gradient integrals. Differentiating $P_{\text{no-fa}}(\rho)$ of (10) under the integral sign — the integrand and its d' -derivatives are bounded, so dominated convergence applies

— gives two one-dimensional Gauss–Hermite integrals,

$$I_c(b_c, b_u; \rho, N) := \int_{-\infty}^{\infty} \frac{1}{\sqrt{1-\rho}} \varphi\left(\frac{b_c - \sqrt{\rho} z}{\sqrt{1-\rho}}\right) \Phi\left(\frac{b_u - \sqrt{\rho} z}{\sqrt{1-\rho}}\right)^{N-1} \varphi(z) dz, \quad (27)$$

$$I_u(b_c, b_u; \rho, N) := \int_{-\infty}^{\infty} \frac{1}{\sqrt{1-\rho}} \Phi\left(\frac{b_c - \sqrt{\rho} z}{\sqrt{1-\rho}}\right) \Phi\left(\frac{b_u - \sqrt{\rho} z}{\sqrt{1-\rho}}\right)^{N-2} \varphi\left(\frac{b_u - \sqrt{\rho} z}{\sqrt{1-\rho}}\right) \varphi(z) dz. \quad (28)$$

At $\rho = 0$ these reduce to the independent products

$$I_c^0 = \varphi(b_c) \Phi(b_u)^{N-1}, \quad I_u^0 = \Phi(b_c) \Phi(b_u)^{N-2} \varphi(b_u),$$

and they share the same Gauss–Hermite nodes used to evaluate $P_{\text{no-fa}}(\rho)$ itself.

Boundary first-order condition. The chain-rule expansion of $\partial R / \partial \alpha$ at $\alpha = 1/N^+$ collects the asymmetric criterion booking into

$$\left. \frac{\partial R}{\partial \alpha} \right|_{1/N^+}(\rho) = d'_{\max} f'(1/N) \left[\beta(r) K_c(v; \rho) - \gamma(r) \frac{K_u(v; \rho)}{N-1} \right], \quad (29)$$

with correlation-aware coefficients

$$K_c(v; \rho) = \frac{1}{4} \left[V v \varphi\left(\frac{d'_b}{2} - c_c^*(\rho)\right) + \text{CR} I_c(b_c^*(\rho), b_u^*(\rho); \rho, N) \right], \quad (30)$$

$$K_u(v; \rho) = \frac{1}{4} \left[(1-V) \varphi\left(\frac{d'_b}{2} - c_u^*(\rho)\right) + (N-1) \text{CR} I_u(b_c^*(\rho), b_u^*(\rho); \rho, N) \right], \quad (31)$$

where $b_i^*(\rho) := c_i^*(\rho) + d'_{\text{base}}/2$, $d'_b \equiv d'_{\text{base}}$, and $f'(a) = (1 - f_0) h'(a)$. The change-trial density terms depend on ρ only through $c_i^*(\rho)$; the no-false-alarm terms I_c, I_u carry the full correlation dependence of the quadrature. Both K_c and K_u are positive.

Proposition 3 (Escape threshold under equicorrelation). *Fix $(N, V, v, d'_{\max}, f_0, h, \text{CR}, \rho)$ with $\rho \in [0, 1)$, and let $(c_c^*(\rho), c_u^*(\rho))$ solve (26). Then the value-aware optimum satisfies $\alpha_c^* = 1/N$ for $r \leq r^\dagger(v; \rho)$ and $\alpha_c^* > 1/N$ for $r > r^\dagger(v; \rho)$, with*

$$r^\dagger(v; \rho) = \frac{K_u(v; \rho)}{(N-1) K_c(v; \rho)}. \quad (32)$$

Proof. Setting (29) to zero and dividing out the strictly positive prefactor $d'_{\max} f'(1/N) \gamma(r) > 0$ gives $r K_c(v; \rho) = K_u(v; \rho)/(N-1)$, using $\beta(r)/\gamma(r) = r$ (which holds for every member of the conservation family of Section A.4); solving for r gives (32). The corner maximum at $\alpha = 1/N$ becomes a saddle as r crosses the threshold; a one-sided finite-difference check at $r = r^\dagger(v; \rho) \pm 0.05$ confirms the boundary slope changes sign at every $(v, \rho) \in \{1, 2, 3\} \times \{0, 0.2\}$ tested. $\square$

Recovery at zero correlation. At $\rho = 0$ the integrand of (27) loses its z -dependence and $I_c \rightarrow I_c^0$, $I_u \rightarrow I_u^0$; the criterion optimum (26) reduces to the corresponding optimum at $\rho = 0$, so $K_c(v; 0) = K_c(v)$ and $K_u(v; 0) = K_u(v)$ of (17)–(18), and Proposition 3 reduces to (16). The recovery is structural — the integrals collapse term by term — and the threshold computed from (32) at $\rho = 0$ reproduces Table 3 across $v \in \{1, 2, 3, 5, 8, 10\}$ to within the criterion-grid resolution ($\leq 5 \times 10^{-4}$ in $r^\dagger$).

v	$r^\dagger(v; 0)$	$r^\dagger(v; 0.2)$	$\Delta r^\dagger$	% $\Delta r^\dagger$	sign-match r^* ?
1	0.343	0.354	+0.0103	+3.0%	— (reference)
2	0.168	0.173	+0.0051	+3.0%	✓
3	0.099	0.113	+0.0132	+13.2%	✓
5	0.050	0.058	+0.0077	+15.2%	✓
8	0.022	0.029	+0.0066	+29.7%	✓
10	0.016	0.019	+0.0030	+18.7%	✓

Table 12: Drift of the escape threshold under correlation at the headline cell $(N, V, d'_{\max}, f_0, h) = (4, 0.5, 2, 0.5, \sqrt{\cdot})$, value-coupled correct rejection, additive conservation. $r^\dagger(v; \rho)$ from (32); $\Delta r^\dagger := r^\dagger(v; 0.2) - r^\dagger(v; 0)$. The drift is positive at every cued value, including the value-blind reference $v = 1$: the regime in which the optimum is pinned at uniform allocation *widens* as correlation grows. The sign of $\Delta r^\dagger$ matches the sign of the empirical peak drift Δr^* of Table 5 at all five $v \neq 1$ rows.

Drift under correlation. Evaluating (32) at the headline cell quantifies how the escape threshold moves when correlation is admitted (Table 12). Two features follow. First, $\Delta r^\dagger(v) > 0$ at every v : the lower edge of the active band moves up under correlation across the whole value family, and because the peak of the value-directed-attention benefit cannot lie below the escape threshold, the peak must drift up with it — the behaviour reported in Section 3.2. Second, the absolute drift in the threshold is bounded by ≤ 0.014 across the family, while the empirical peak drift in Table 5 reaches 0.13 at $v = 2$: the peak responds not only to the lower edge but also to the upward drift of the value-blind reference threshold (which widens the band’s upper edge) and to how steeply the optimal allocation climbs above uniform once it escapes. Equation (32) pins the lower-edge drift exactly; a closed form for the peak location itself, which would additionally model the climb of α_c^* above the threshold, is a natural further step. The two gradient integrals (27)–(28) make the concentration-cost channel of Section A.1 analytic on the lower edge: the same one-factor reduction that renders $P_{\text{no-fa}}(\rho)$ tractable renders the boundary slope tractable.

A.3 Symmetric recovery at unit benefit/cost ratio

At $r = 1$ the asymmetric benefit/cost weighting $\beta(r) = 2r/(r+1)$, $\gamma(r) = 2/(r+1)$ collapses to the symmetric special case $\beta = \gamma = 1$, in which a single shared transfer function drives both the cued and the uncued sensitivity. We record the recovery at the strength the model supports: a real-number identity that holds universally, and a bit-exact identity that holds at the validation configuration.

The collapse as a real-number identity. The asymmetric sensitivity rule of Section 2.3 writes $d'_i(\alpha) = d'_{\text{base}} + s_i(r)(d'_{\max} f(\alpha) - d'_{\text{base}})$ with $s_i \in \{\beta, \gamma\}$ according to whether location i is attended; at $r = 1$ both slopes are unit and the expression collapses to $d'_i(\alpha) = d'_{\max} f(\alpha)$, the symmetric rule. Because the downstream signal-detection, expected-reward, and criterion-optimisation stack is a deterministic function of the d' arrays alone, identical arrays force identical optima.

Proposition 4 (Real-number recovery at unit ratio). *For every $(V, v, N, d'_{\max}, f_0, h)$ in the model’s domain and at any correlation $\rho \in [0, 1)$, the asymmetric model evaluated at $r = 1$ produces the same d' arrays as the symmetric special case $\beta = \gamma = 1$. Consequently the optimal cued share, the optimal cued and uncued criteria, the optimal reward, and the criterion fraction all match the symmetric baseline as real-number identities.*

This is a property of the algebra rather than of the numerical implementation: the cancellation $d'_{\text{base}} - d'_{\text{base}} = 0$ is the entire content of the recovery.

Bit-exact identity at the validation configuration. A sharper statement holds at the configuration used to validate the model. The asymmetric code path at $r = 1$ forms $\text{fl}(a + \text{fl}(1.0 \cdot \text{fl}(x - a)))$ with $a = d'_{\text{base}}$ and $x = d'_{\text{max}} f(\alpha)$. Three facts compose: (i) $\beta(1)$ and $\gamma(1)$ are the exact binary float 1.0, so the multiplicative slope round-trips identically; (ii) Sterbenz’s lemma [17, 18] guarantees $\text{fl}(x - a) = x - a$ exactly whenever $a/2 \leq x \leq 2a$; and (iii) at the validation configuration $(N, d'_{\text{max}}, f_0, h) = (4, 2.0, 0.5, \sqrt{\cdot})$, value-coupled correct rejection, the baseline is $a = d'_{\text{max}}(f_0 + (1 - f_0)h(1/N)) = 1.5$ and the swept range $x = d'_{\text{max}} f(\alpha) \in [1.0, 2.0]$ lies inside the Sterbenz band $[a/2, 2a] = [0.75, 3.0]$ at every grid point.

Proposition 5 (Bit-exact symmetric recovery). *Under value-coupled correct rejection at the validation configuration, the asymmetric implementation evaluated at $r = 1$ returns d' arrays bit-identical to the symmetric implementation; hence the optimal cued share and optimal reward match it to zero difference, not merely to machine epsilon.*

The hypothesis of Proposition 5 is sufficient but not necessary, and fails once the smallest swept output $d'_{\text{max}} f_0$ drops below $d'_{\text{base}}/2$, i.e. once

$$f_0 < \frac{h(1/N)}{1 + h(1/N)} = \frac{1}{3} \quad (h = \sqrt{\cdot}, N = 4). \quad (33)$$

Off-band, bit-identity is lost and the two paths drift at the order of a unit in the last place. The universal statement is therefore that the symmetric recovery is exact to machine precision; the bit-for-bit identity is a structural guarantee of the chosen configuration rather than a property of the model.

Unit ratio is the smooth centre of the family. The collapse at $r = 1$ is the interior of a smooth, two-sided family, not a removable singularity. The slope ratio $\beta(r)/\gamma(r) = r$ is continuous and differentiable through $r = 1$, with $d\beta/dr|_1 = 1/2$ and $d\gamma/dr|_1 = -1/2$. A continuity check at $r = 1 \pm \epsilon$ confirms that the maximum reward difference against the symmetric model scales linearly in $|r - 1|$, with slope ≈ 0.084 reward units per unit shift. Unit ratio is the balanced point of the enhancement-versus-suppression trade-off (Definition 1), the natural null about which the value-directed-attention effects of Section 3 are organised. The recovery is invariant to the conservation rule: at $r = 1$ the ratio identity $\beta/\gamma = 1$ forces $\beta = \gamma$, and any conservation constraint in the family of Section A.4 then forces $\beta = \gamma = 1$, so Propositions 4–5 hold under every conservation order without modification (Corollary 7).

A.4 The power-mean conservation family

The benefit/cost weights satisfy the asymmetry ratio $\beta/\gamma = r$ together with a conservation constraint; the main text uses the additive rule $\beta + \gamma = 2$ (Eq. 4). This subsection embeds that rule in a one-parameter family, derives the family’s monotonicity, and establishes the two invariances on which the escape threshold and the symmetric recovery rest.

The family and its closed-form weights. For $p \in \mathbb{R}$, the power (Hölder) mean of order p on positive reals (x, y) is

$$M_p(x, y) = \begin{cases} \left(\frac{x^p + y^p}{2} \right)^{1/p}, & p \neq 0, \\ \sqrt{x y}, & p = 0, \end{cases} \quad (34)$$

with limiting cases $M_{-\infty} = \min$ and $M_{+\infty} = \max$ [19]. The conservation family is the constraint $M_p(\beta, \gamma) = 1$ together with $\beta/\gamma = r$. Substituting $\beta = r\gamma$ for $p \neq 0$ gives the closed-form weights

$$\boxed{\gamma(r; p) = \left(\frac{2}{r^p + 1} \right)^{1/p}, \quad \beta(r; p) = r \gamma(r; p),} \quad (35)$$

with the geometric-mean limit $\gamma(r;0) = 1/\sqrt{r}$, $\beta(r;0) = \sqrt{r}$. The case $p = 1$ recovers the additive rule $\beta = 2r/(r+1)$, $\gamma = 2/(r+1)$; $p = 0$ is the multiplicative (geometric-mean) endpoint $\beta\gamma = 1$. Three identities hold at every p : $\beta/\gamma = r$ by construction; $\beta(1;p) = \gamma(1;p) = 1$; and $\text{sgn}(\beta - \gamma) = \text{sgn}(r - 1)$.

Monotonicity in the conservation order. The classical power-mean inequality makes $p \mapsto M_p(x, y)$ strictly increasing for fixed $x \neq y$ [19]. Under the constrained form $M_p(\beta, \gamma) = 1$, this translates into an exact identity on $\partial \ln \gamma / \partial p$. Writing $L(p) = \ln \gamma(r; p) = (1/p) \ln(2/(r^p + 1))$ and $\theta_p := r^p/(r^p + 1)$, direct differentiation gives

$$\boxed{\frac{\partial}{\partial p} \ln \gamma(r; p) = -\frac{1}{p^2} D_{\text{KL}}(\text{Bern}(\theta_p) \parallel \text{Bern}(1/2))}, \quad (36)$$

where the bracket of the expansion, $\ln(2/(r^p + 1)) + p \theta_p \ln r$, equals $\theta_p \ln(2\theta_p) + (1 - \theta_p) \ln(2(1 - \theta_p)) = D_{\text{KL}}(\text{Bern}(\theta_p) \parallel \text{Bern}(1/2))$. Since $D_{\text{KL}} \geq 0$ with equality only at $\theta_p = 1/2$ (i.e. $r = 1$), (36) yields, for every $r \neq 1$ and every $p \neq 0$,

$$\frac{\partial \gamma}{\partial p}(r; p) < 0, \quad \frac{\partial \beta}{\partial p}(r; p) = r \frac{\partial \gamma}{\partial p} < 0. \quad (37)$$

Continuity through $p = 0$ holds by series expansion, with $\lim_{p \rightarrow 0} \partial \ln \gamma / \partial p = -(\ln r)^2/8$. Both weights therefore increase strictly as the conservation order decreases, at every $r \neq 1$. A finite-difference cross-check of (36) across a range of (r, p) pairs agrees to $\leq 1.5 \times 10^{-10}$.

The symmetric corner and the recovery at unit ratio.

Proposition 6 (Symmetric-corner identity). *For every $p \in \mathbb{R}$ (including $p = 0$), $\beta(1; p) = \gamma(1; p) = 1$.*

Proof. At $r = 1$, $r^p = 1$ for every p , so (35) gives $\gamma = (2/2)^{1/p} = 1$ and $\beta = 1$ for $p \neq 0$; the geometric-mean limit gives $\gamma = 1/\sqrt{1} = 1$, $\beta = 1$ at $p = 0$. $\square$

Corollary 7 (Conservation-form invariance of the symmetric recovery). *The symmetric recovery of Proposition 4 holds at every conservation order: at $r = 1$ and any p , the asymmetric d' -map collapses pointwise to the symmetric baseline, $d'_c(\alpha; 1, p) = d'_u((1 - \alpha)/(N - 1); 1, p) = d'_{\max} f(\alpha)$.*

Proof. By Proposition 6, $\beta(1; p) = \gamma(1; p) = 1$ at every p , so the d' -map at $r = 1$ reads $d'_c(\alpha; 1, p) = d'_{\text{base}} + 1 \cdot (d'_{\max} f(\alpha) - d'_{\text{base}}) = d'_{\max} f(\alpha)$ with no p -dependence; the uncued counterpart is identical. The optima then coincide with the symmetric baseline as in Proposition 4. $\square$

Conservation-form invariance of the escape threshold.

Proposition 8 (Escape threshold is conservation-form invariant). *The escape threshold $r^\dagger(v) = K_u(v)/[(N - 1)K_c(v)]$ of (16) is independent of the conservation order p .*

Proof. **(Step 1)** At the uniform allocation $\alpha = 1/N$ the d' -map reads $d'_c(1/N; r, p) = d'_{\text{base}} + \beta(r; p)(d'_{\max} f(1/N) - d'_{\text{base}}) = d'_{\text{base}}$, because the bracket vanishes by the definition $d'_{\text{base}} := d'_{\max} f(1/N)$. The vanishing bracket annihilates any value of $\beta(r; p)$, so d'_c at uniform allocation is p -independent; the uncued per-location allocation is also $1/N$, so $d'_u = d'_{\text{base}}$ likewise. The pair $(d'_{\text{base}}, d'_{\text{base}})$ is therefore independent of p (and of r). **(Step 2)** At this fixed allocation the criterion-optimal pair (c_c^*, c_u^*) is the argmax of a reward functional whose only p -dependence entered through (d'_c, d'_u) ; by Step 1 those are p -independent, so the criteria are too. **(Step 3)** The gradient coefficients $K_c(v), K_u(v)$ are the sensitivity partials of the reward evaluated at the p -independent $(d'_{\text{base}}, d'_{\text{base}}, c_c^*, c_u^*)$, and the functional form depends only on (V, v, N, CR) ; hence K_c, K_u and their ratio $r^\dagger(v)$ are p -independent. $\square$

The escape threshold is thus an algebraic invariant of the family: the left edge of the active band is fixed across conservation orders, while the peak inside the band — set at allocations $\alpha \neq 1/N$, where the gain bracket no longer vanishes — varies with p and is reported as a band in Section 3.2.

The criterion fraction under the conservation order. The dependence of the criterion fraction on p admits a chain-rule sign analysis. With $CF = (R_{P3} - R_{P1})/(R_{P4} - R_{P1})$ (Eq. 15) and the envelope theorem applied at each policy’s own optimum, the p -partials thread through the d' -map:

$$\frac{\partial R_{P1}}{\partial p} = \left. \frac{\partial R_{P1}}{\partial d'_c} \right|_{\star} \frac{\partial d'_c}{\partial p} + \left. \frac{\partial R_{P1}}{\partial d'_u} \right|_{\star} \frac{\partial d'_u}{\partial p}, \quad (38)$$

with

$$\frac{\partial d'_c}{\partial p} = \frac{\partial \beta}{\partial p} (d'_{\max} f(\alpha) - d'_{\text{base}}), \quad \frac{\partial d'_u}{\partial p} = \frac{\partial \gamma}{\partial p} (d'_{\max} f((1 - \alpha)/(N - 1)) - d'_{\text{base}}). \quad (39)$$

By (37), both $\partial \beta / \partial p$ and $\partial \gamma / \partial p$ are negative for $r \neq 1$; for $\alpha \geq 1/N$ the first bracket of (39) is non-negative and the second non-positive, so the cued and uncued sensitivity channels move with opposite sign in p .

Proposition 9 (Criterion-only reward is invariant at uniform allocation). *At the criterion-only policy P3, fixed at $\alpha = 1/N$, $\partial R_{P3} / \partial p \equiv 0$.*

Proof. By Step 1 of Proposition 8 the brackets in (39) vanish at $\alpha = 1/N$, so $\partial d'_c / \partial p = \partial d'_u / \partial p = 0$ and (38) gives $\partial R_{P3} / \partial p = 0$. $\square$

Using Proposition 9, the criterion-fraction gradient simplifies to

$$\frac{\partial CF}{\partial p} = \frac{-(R_{P4} - R_{P1}) \partial_p R_{P1} - (R_{P3} - R_{P1}) (\partial_p R_{P4} - \partial_p R_{P1})}{(R_{P4} - R_{P1})^2},$$

whose sign reduces to a competition between $-\partial_p R_{P1} \geq 0$ and $-(\partial_p R_{P4} - \partial_p R_{P1})$. The uniform inequality $|\partial_p R_{P4}| \leq |\partial_p R_{P1}|$ would make $\partial CF / \partial p \leq 0$ pointwise; this holds empirically across the full parameter sweep, where swapping the additive rule ($p = 1$) for the multiplicative endpoint ($p = 0$) gives $\Delta CF \leq 0$ in every one of the 4,410 valid cells, with no reverse flips, doubling the fraction of cells with $CF < 0.5$ from 4.0% to 8.3% while moving the median by less than 0.005. A closed-form proof of the uniform inequality is left open: both $\partial_p R_{P1}$ and $\partial_p R_{P4}$ carry the moving optimum $\alpha^*(p)$ that the chain rule does not collapse.

Scope. The conservation-order analysis is carried at zero correlation; the joint dependence on (p, ρ) would compose the power-mean gradient (36) with the correlation gradients (27)–(28) and is a natural further step. The analysis assumes a single global benefit/cost ratio and homogeneous uncued allocation; relaxing either to per-location heterogeneity or to the full N -dimensional allocation simplex leaves the conservation weights (β, γ) entering the per-location d' -map unchanged, so the two relaxations compose with the conservation order.

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